04

时序逻辑电路（下）



4 - 1 施密特触发器

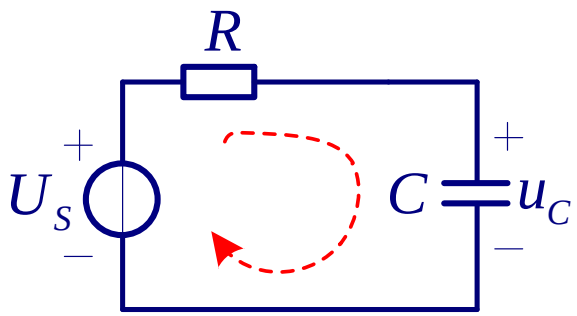
4 - 2 单稳态触发器

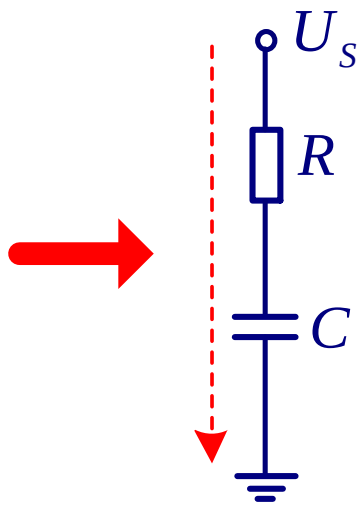
4 - 3 多谐振荡器

4 - 4 555定时器

预备知识

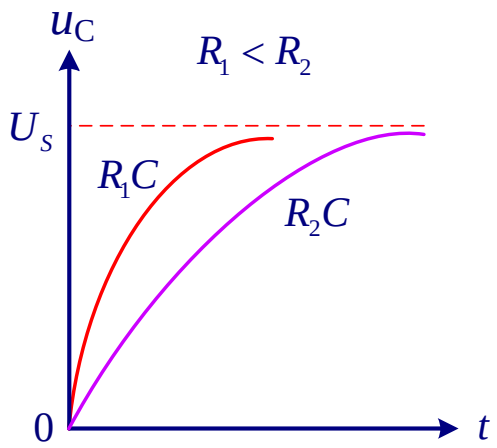
(1) 电容充电



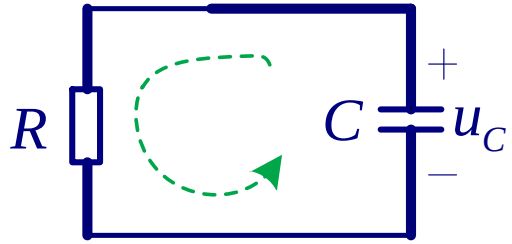


$$\begin{cases} RC\dfrac{du_C}{dt}+u_C=U_S \\ u_C(0)=0 \end{cases}$$

$$u_C(t)=U_S-U_Se^{-\frac{t}{RC}}$$

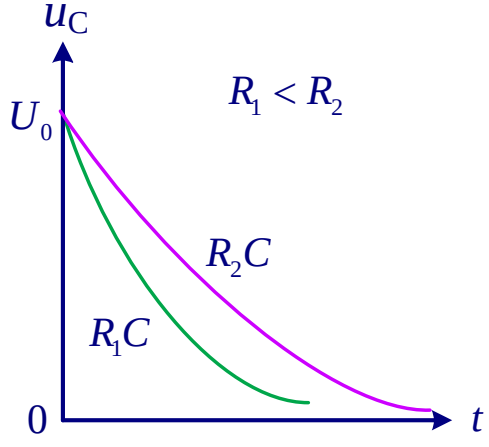


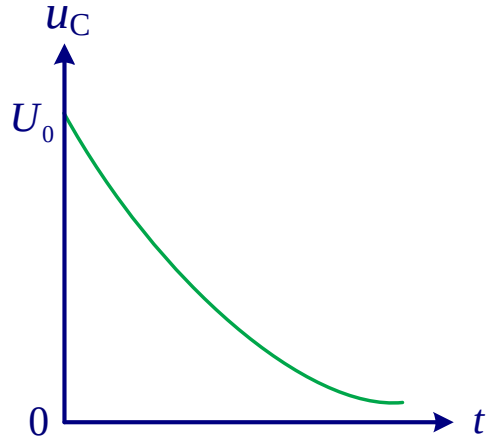
(2) 电容放电



$$\begin{cases} RC\dfrac{du_C}{dt}+u_C=0 \\ u_C(0)=U_0 \end{cases}$$

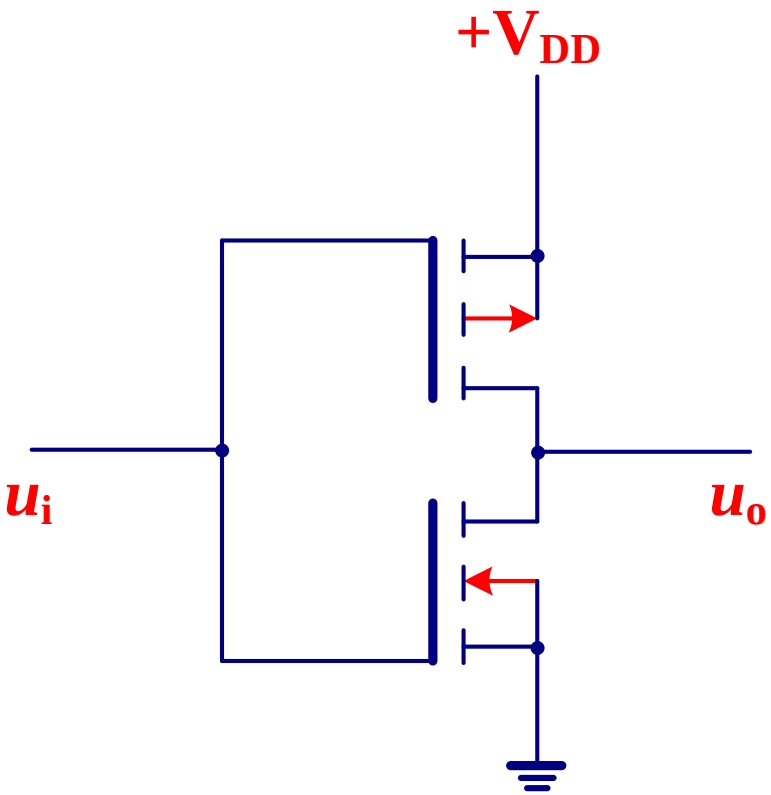
$$u_C(t)=U_0e^{-\frac{t}{RC}}$$

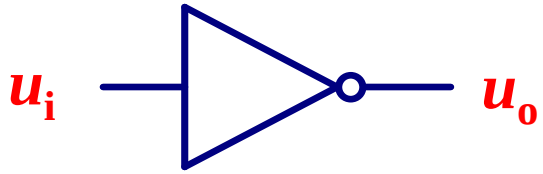


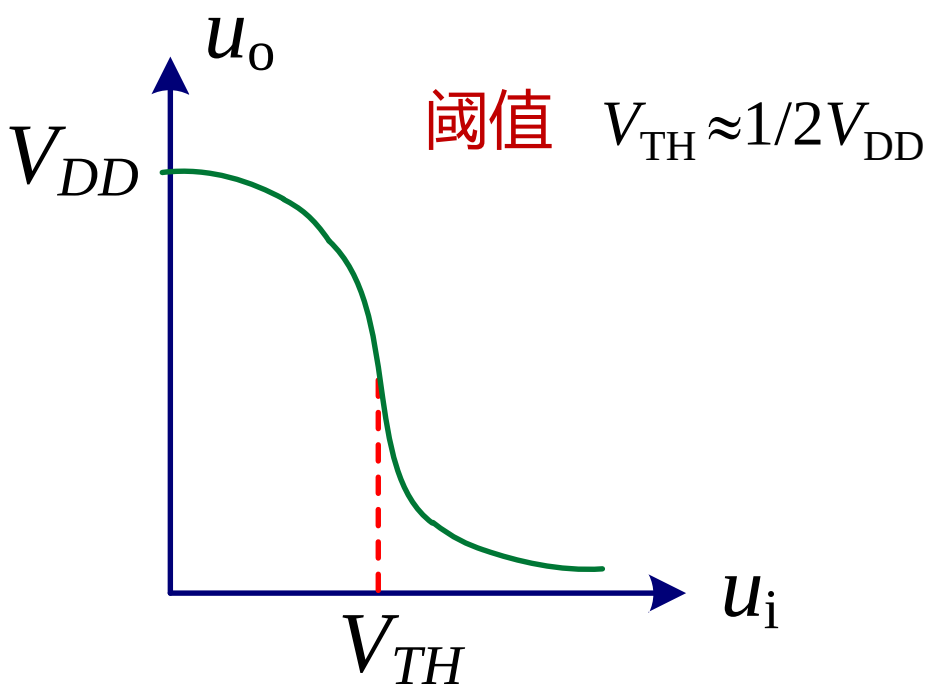


(3) 阈值电压

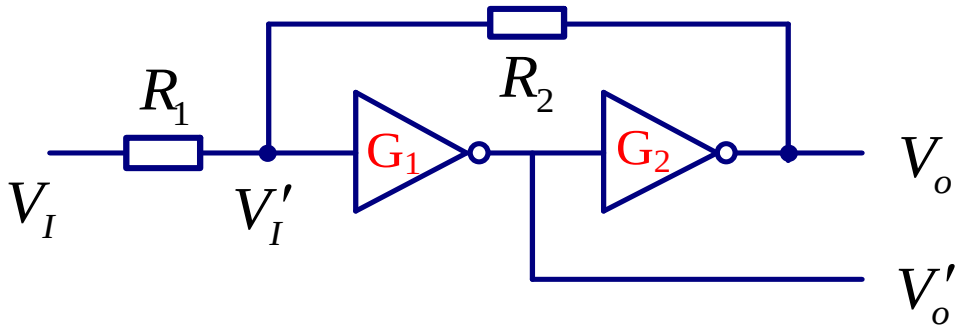
CMOS Inverter







1、施密特触发器 Schmitt-Trigger



G_1、G_2 的 $V_{TH} \approx 1/2V_{DD}$

$R_1 < R_2$

$$V_I' = \frac{\frac{V_I}{R_1} + \frac{V_o}{R_2}}{\frac{1}{R_1} + \frac{1}{R_2}} = \frac{R_2 V_I + R_1 V_o}{R_1 + R_2} = V_{TH}$$

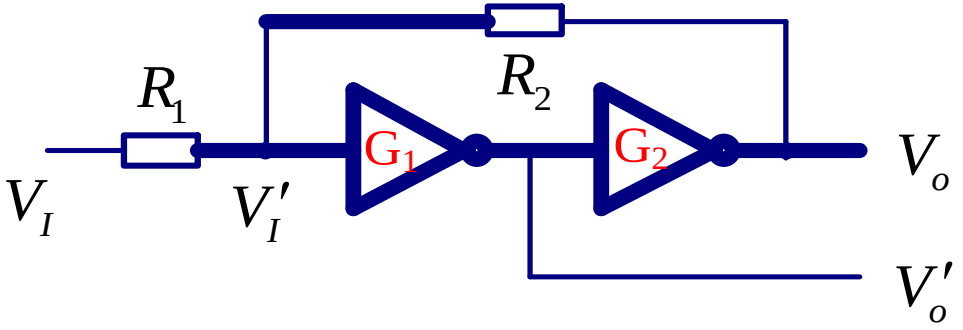
Positive-going threshold voltage

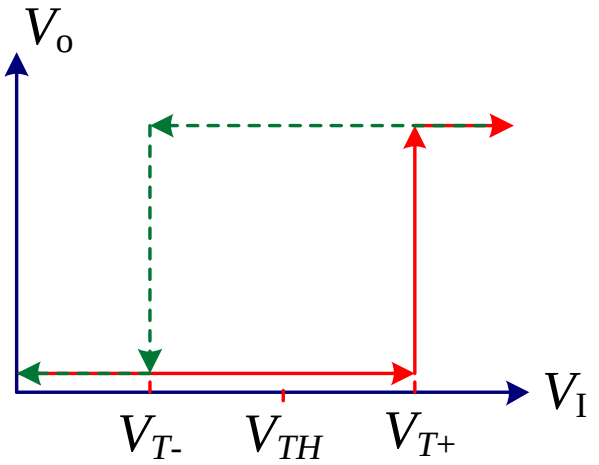
$$V_{TH} = \frac{R_2 V_{T+} + R_1 \times 0}{R_1 + R_2} \Rightarrow V_{T+} = \left(1 + \frac{R_1}{R_2}\right) V_{TH}$$

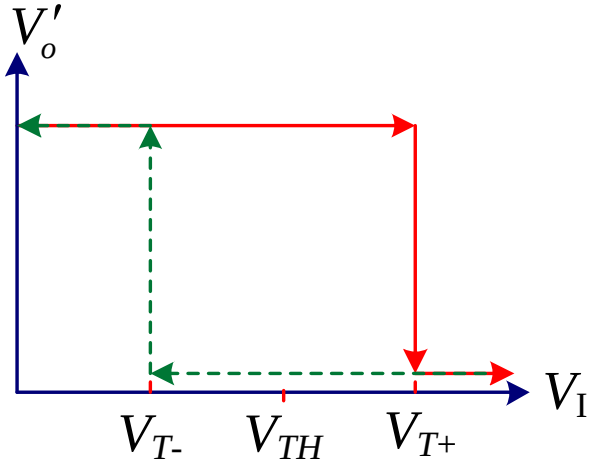
Negative-going threshold voltage

$$V_{TH} = \frac{R_2 V_{T-} + R_1 \times V_{DD}}{R_1 + R_2} \Rightarrow V_{T-} = \left(1 - \frac{R_1}{R_2}\right) V_{TH}$$

a special type of bi-stable device that has two threshold voltages





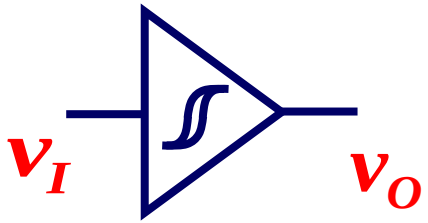


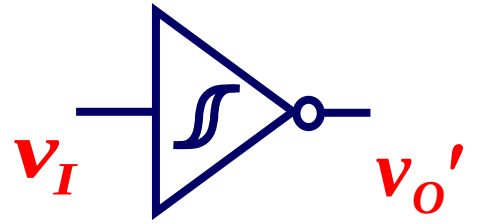
$$V_{T+}=\left(1+\frac{R_1}{R_2}\right)V_{TH}$$

$$V_{T-}=\left(1-\frac{R_1}{R_2}\right)V_{TH}$$

输入输出同相

反相输出



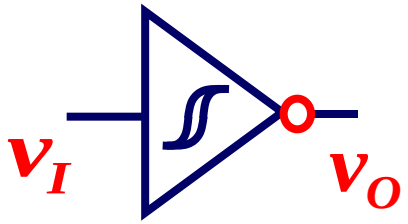


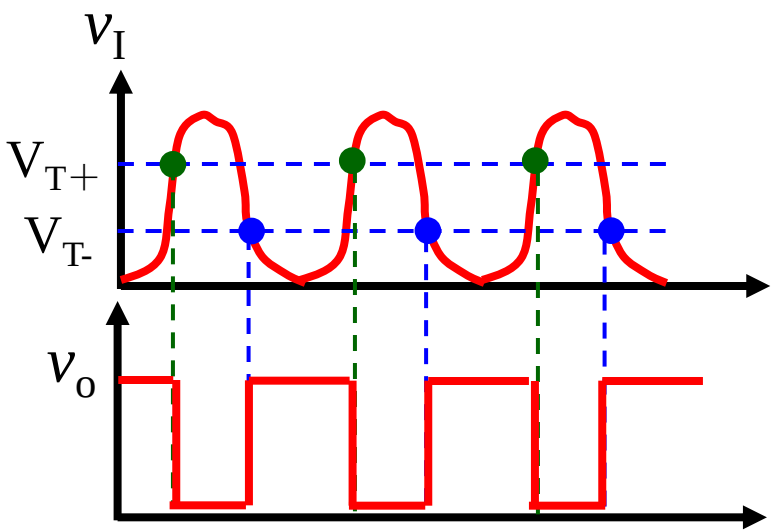
回差电压 $\Delta V_T = V_{T+} - V_{T-} = 2\dfrac{R_1}{R_2}V_{TH}$

SJTU

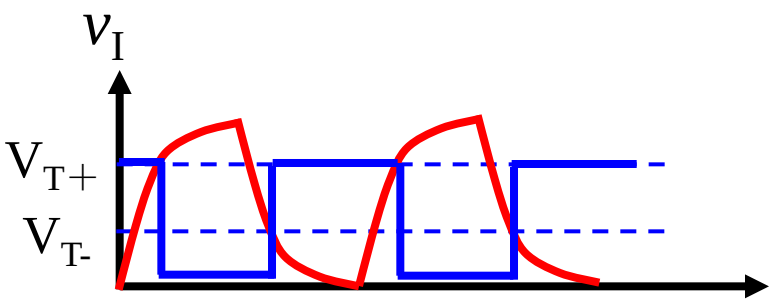
施密特触发器的应用

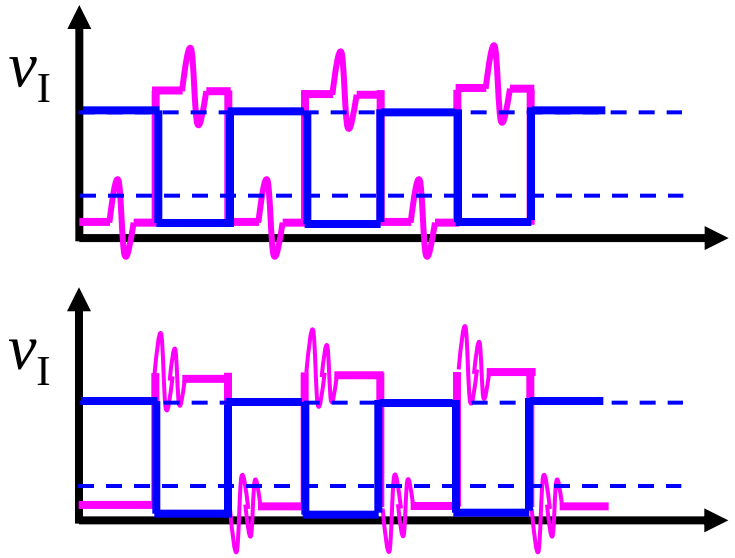
(1) 波形变换



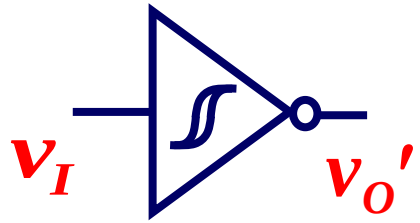


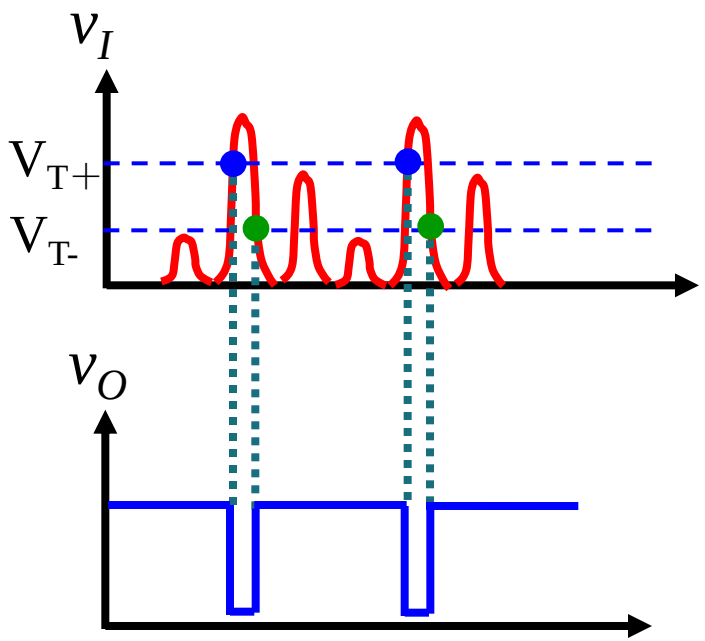
(2) 脉冲整形



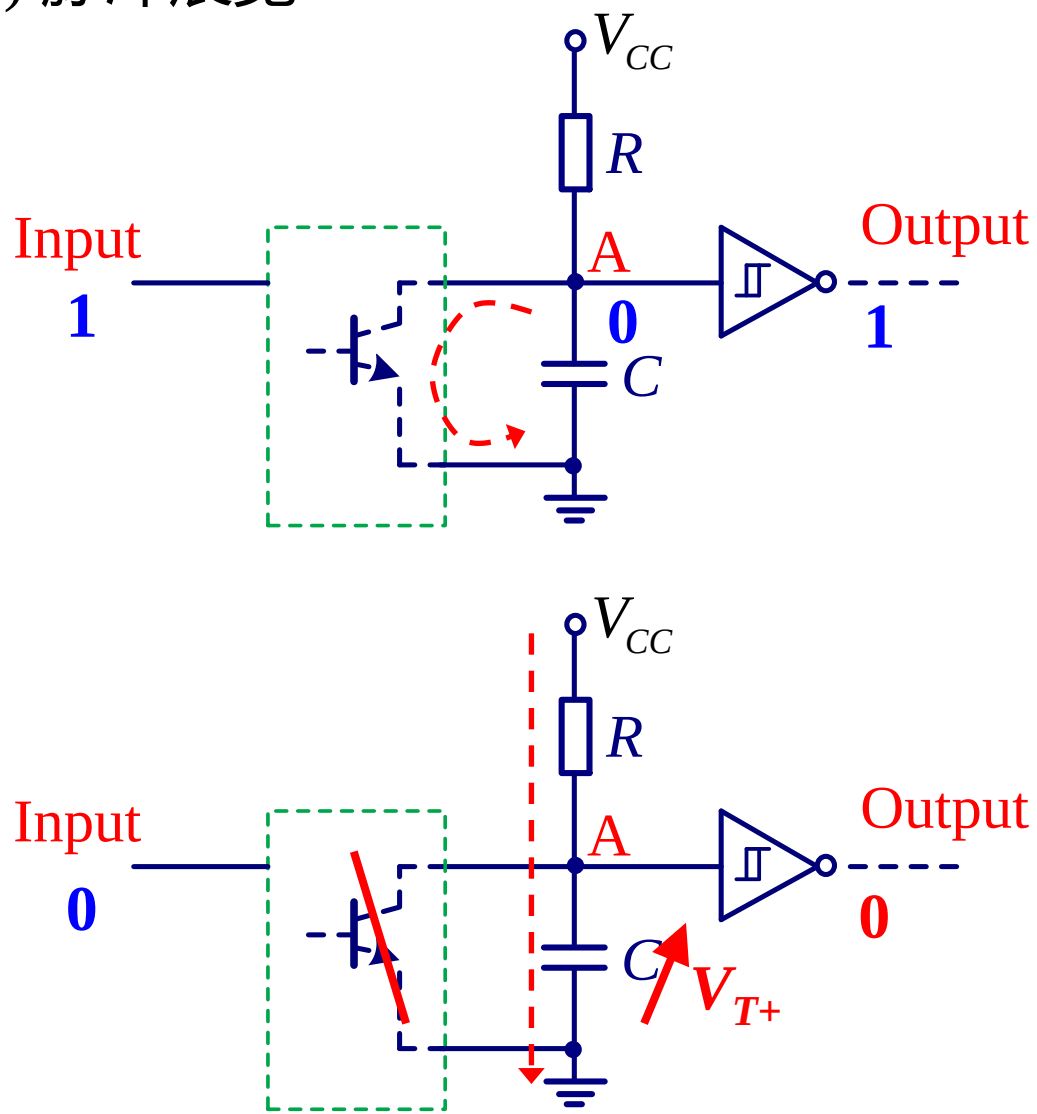


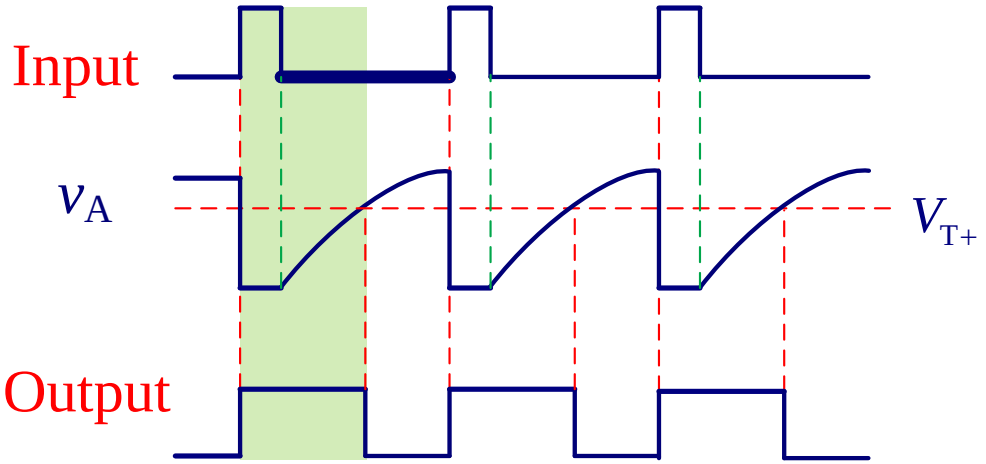
(3) 脉冲鉴幅





(4) 脉冲展宽


V_{CC}
R
Input
1
A
0
Output
1
C
V_{CC}
R
Input
0
A
Output
0
C
V_{T+}


Input
v_A
V_{T+}
Output

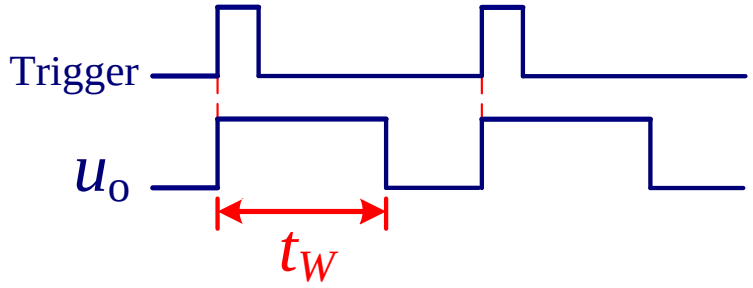
2、单稳态触发器 One-shots　　Monostable

特点：

(1) 电路有一个稳态、一个暂稳态；

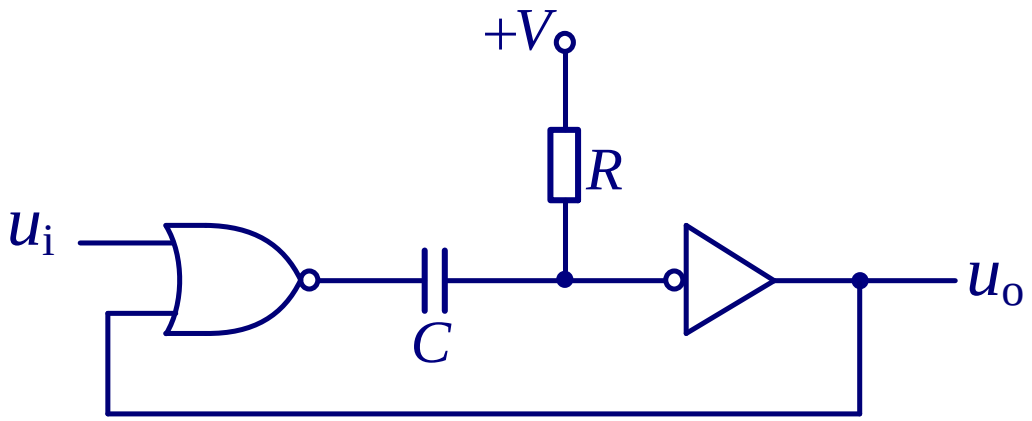
(2) 在外来触发信号作用下，电路由稳态翻转到暂稳态；暂稳态不能长久保持，经过一段时间电路会自动返回稳态。

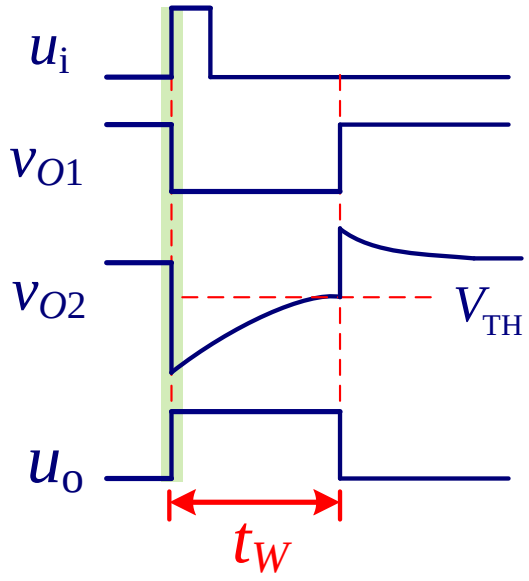
(3) 暂稳态持续的时间取决于电路本身。



应用：广泛应用于脉冲整形、延时以及定时等。

A Simple One-Shot

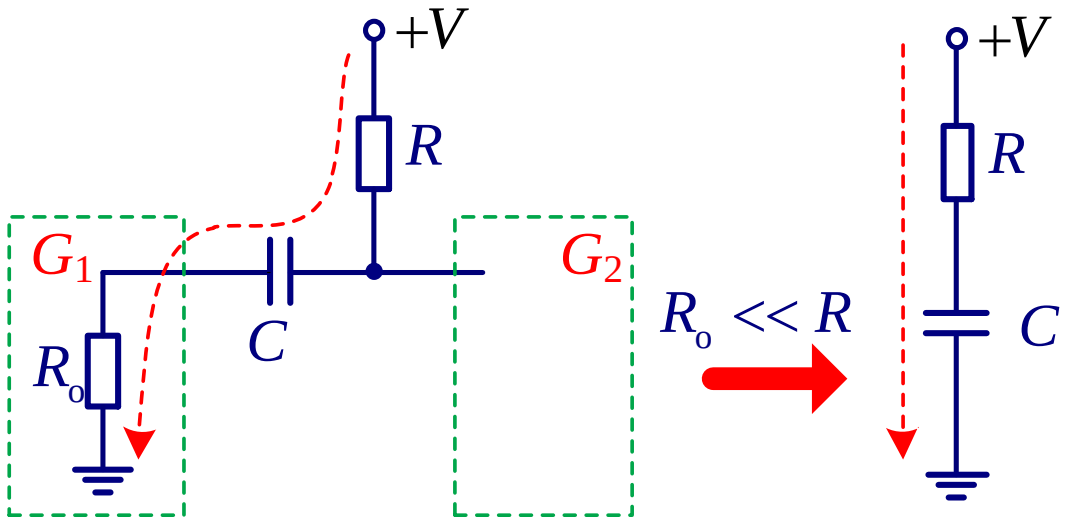




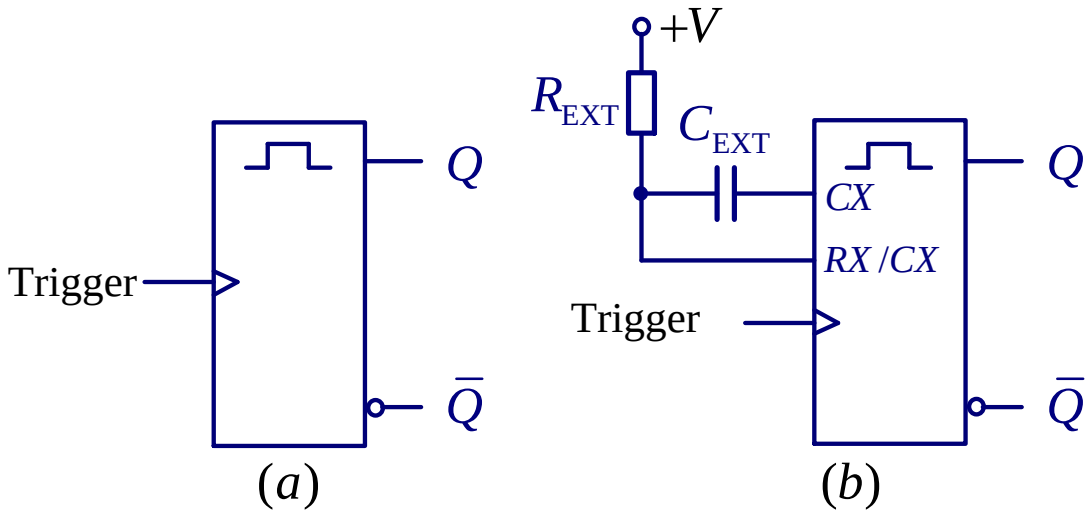
$$t_W = RC\ln 2 \approx 0.7RC$$

$$\left[\because V_{TH} = V(1-e^{-\frac{t}{RC}}) = \frac{V}{2}\right.$$

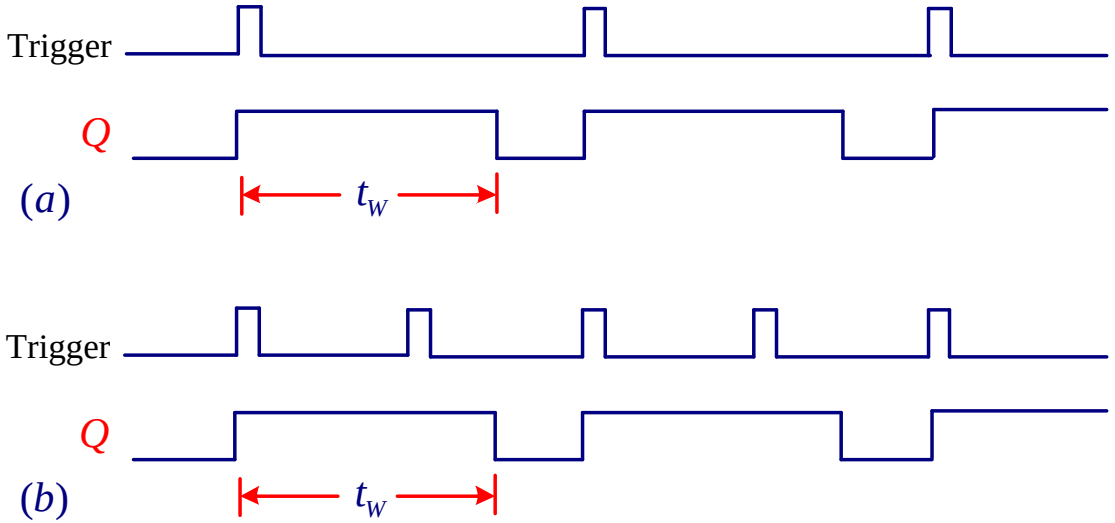
$$\left.\therefore e^{\frac{t_w}{RC}} = 2 \text{ 则 } t_w = RC\ln 2\right]$$

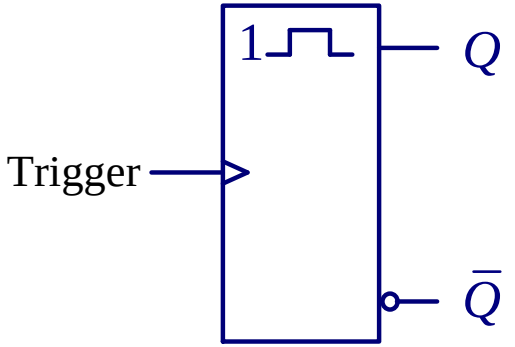


Basic One-shot Logic Symbols

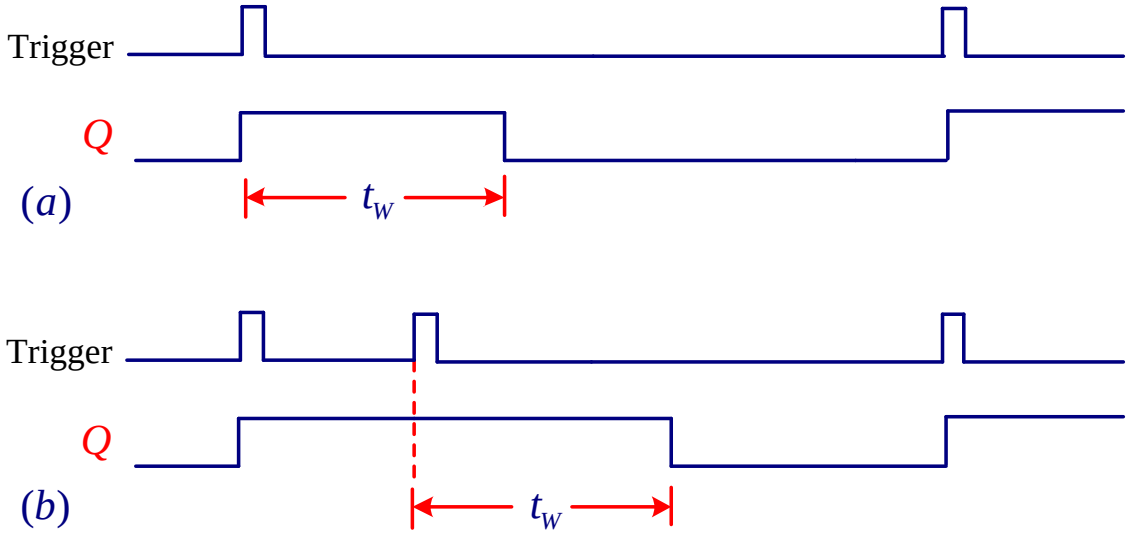


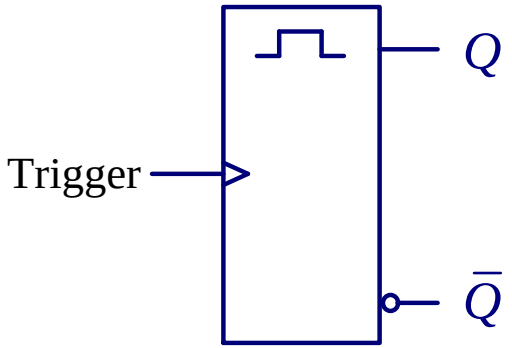
Nonretriggerable One-shot



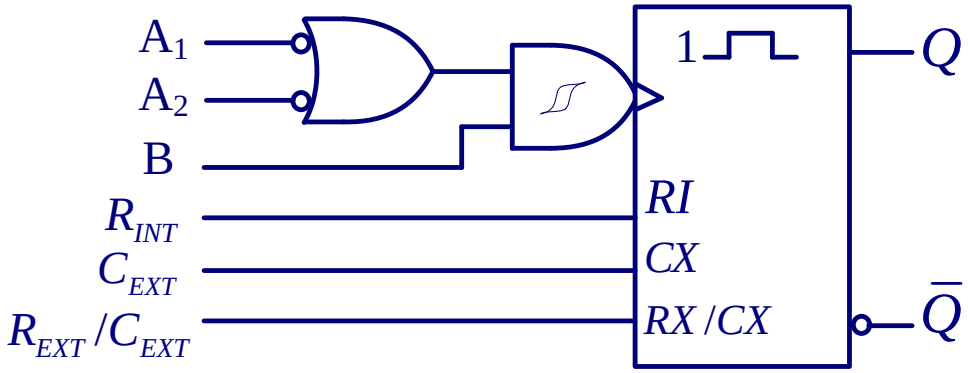


Retriggerable One-shot

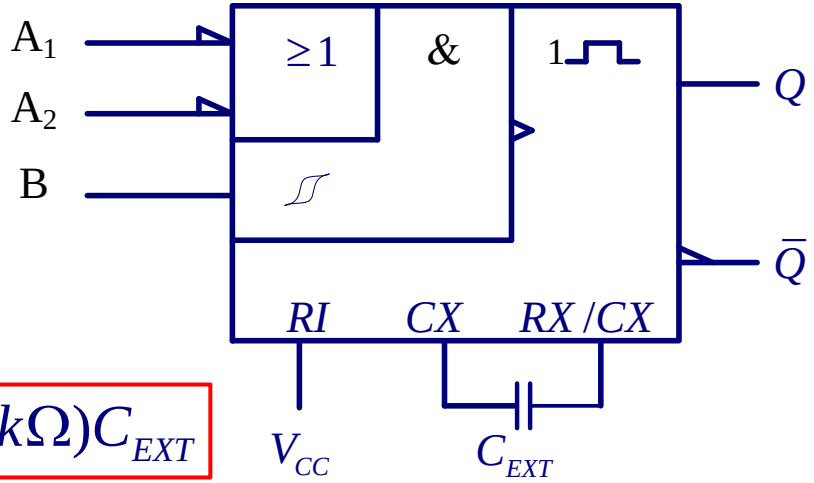




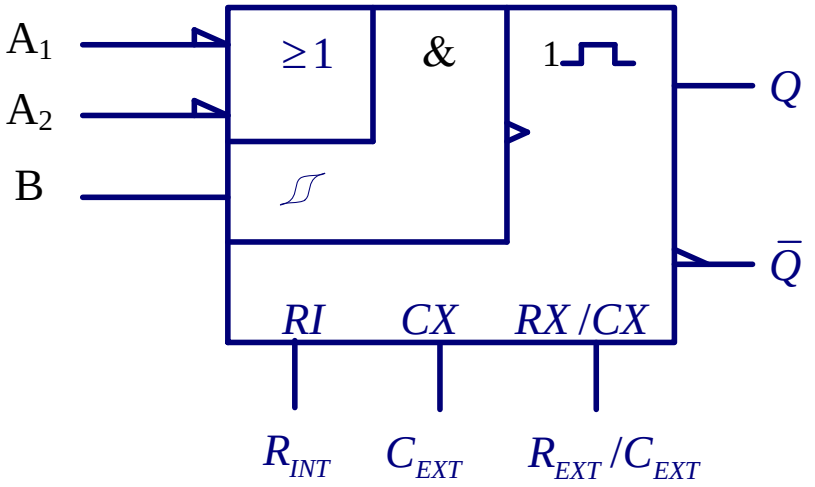
74121: Nonretriggerable One-shot

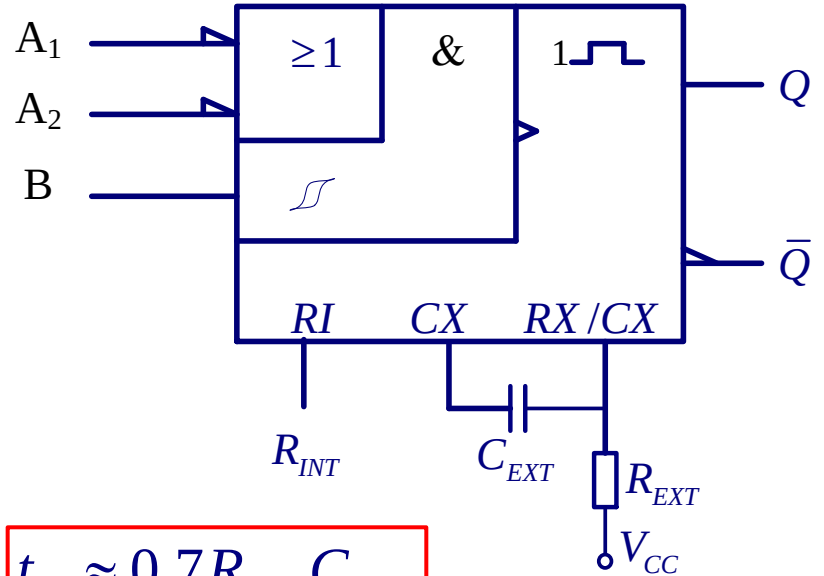


$t_W \approx 30ns$



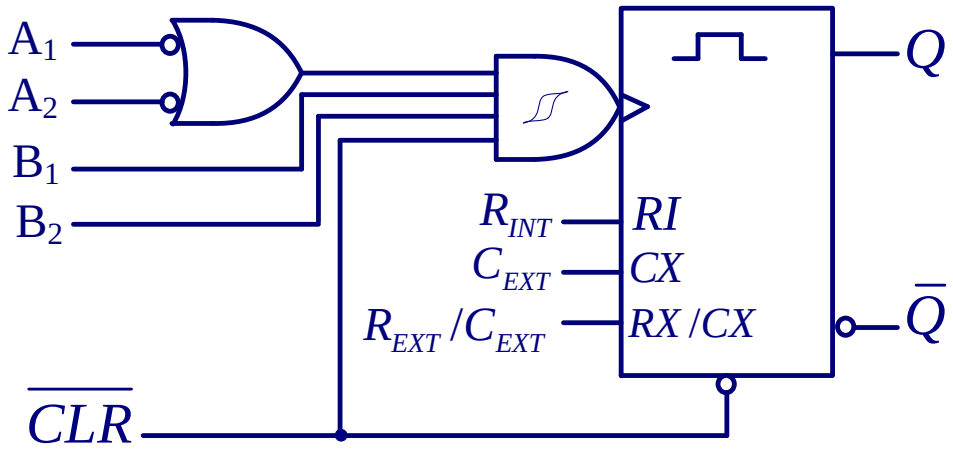
$t_W \approx 0.7(2k\Omega)C_{EXT}$

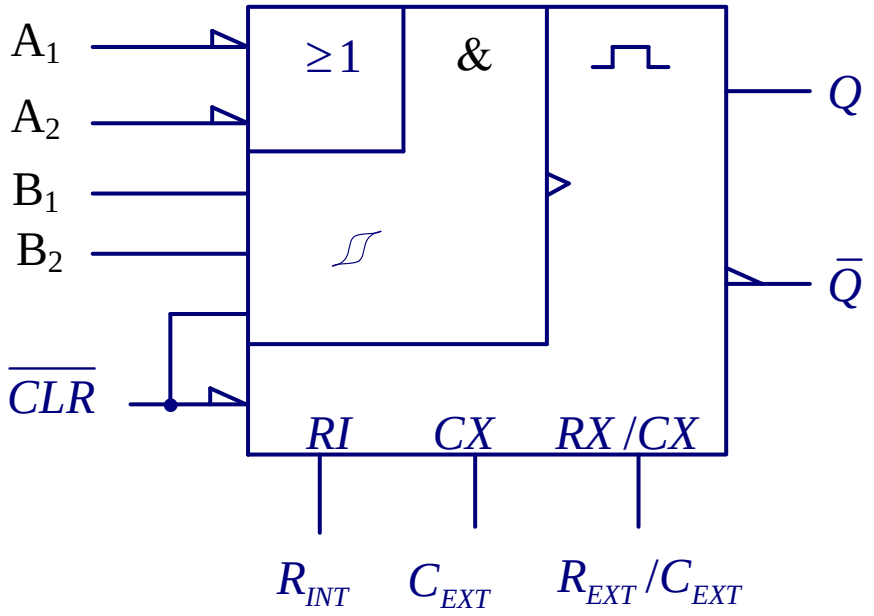




$t_W \approx 0.7R_{EXT}C_{EXT}$

74122: Retriggerable One-shot

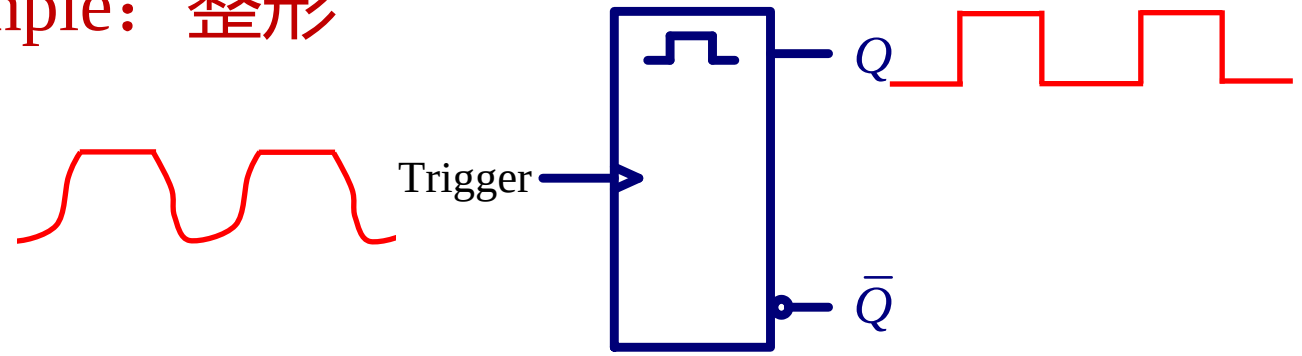




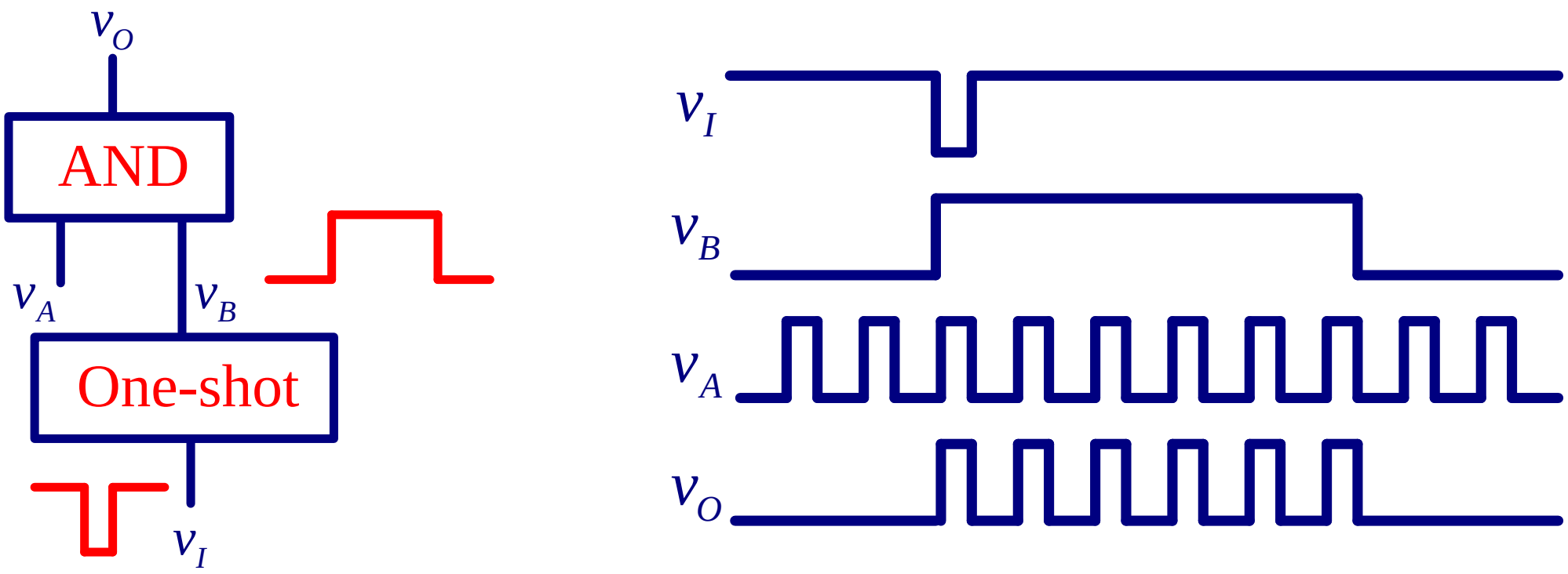
$t_W \approx 45ns$ no external resistor and capacitor

$$t_W = 0.32RC_{EXT}\left(1+\frac{0.7}{R}\right)ns$$ With external resistor and capacitor

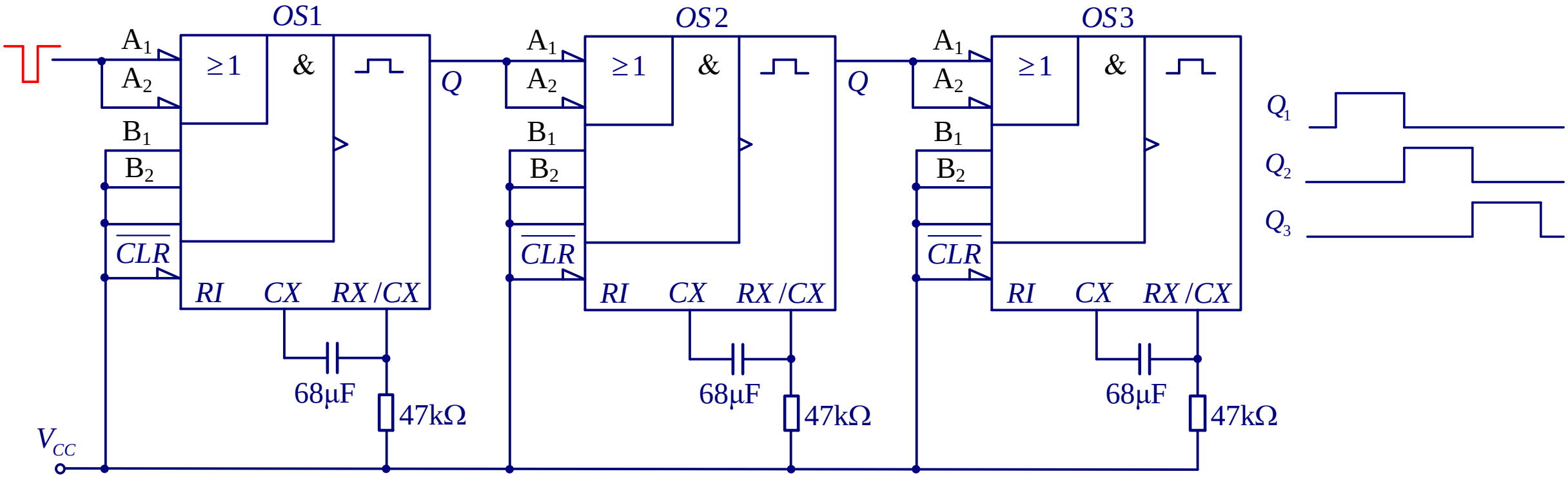
Application example：整形



Application example：定时



Application example： Sequential Pulse Generator



构成一系列定时器，可以用来点亮一系列电灯，形成提示信号如：车辆转向信号。

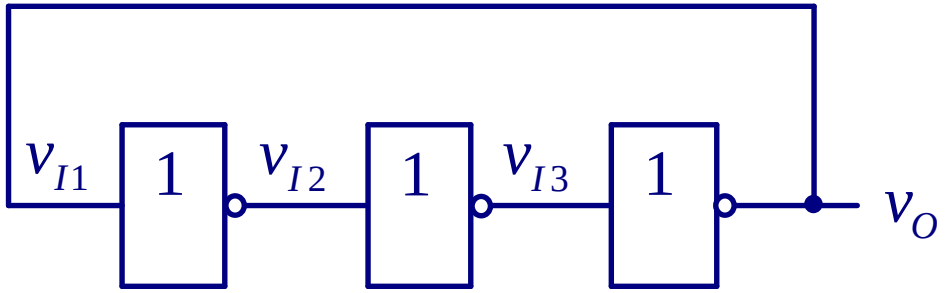
$$t_W = 0.32 \times 47 \times 10^3 \times 68 \times 10^{-6} \times \left(1 + \frac{0.7}{47 \times 10^3}\right) \approx 1s$$

3、多谐振荡器 Multivibrator Astable

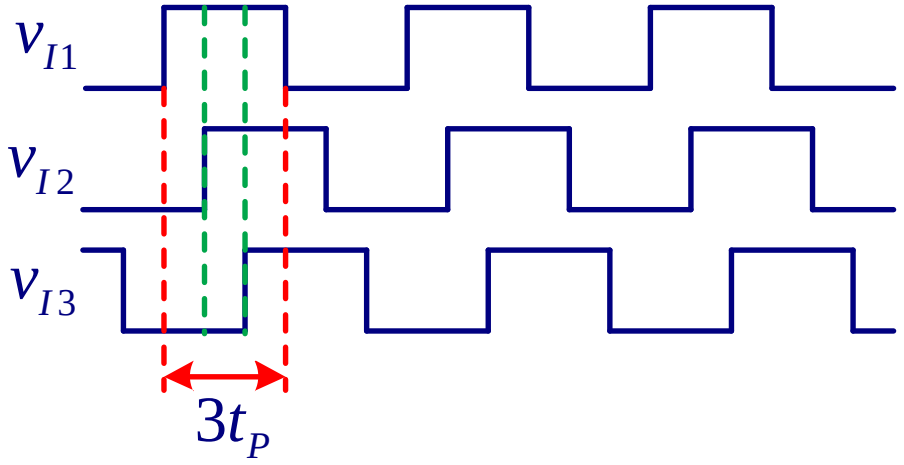
多谐振荡器是一种自激振荡电路，接通电源后无需外接触发信号就能产生一定频率和幅值的矩形脉冲或方波。由于矩形脉冲中含有丰富的高次谐波分量，所以习惯上把矩形波振荡器叫多谐振荡器

多谐振荡器工作过程中不存在稳定状态，所以又称无稳态电路。

A) 环形多谐振荡器



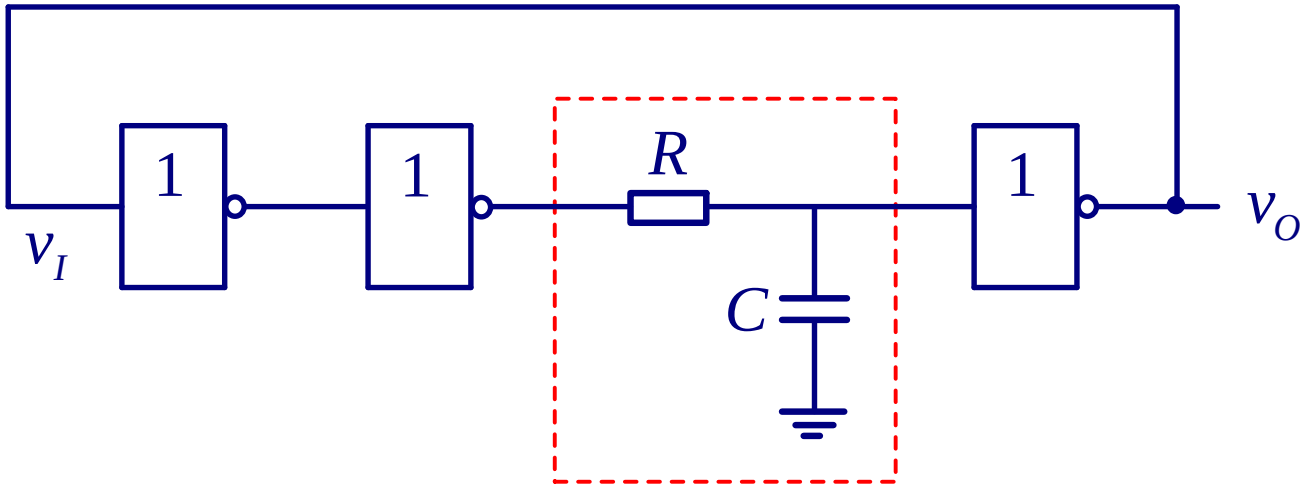
利用延迟负反馈产生振荡



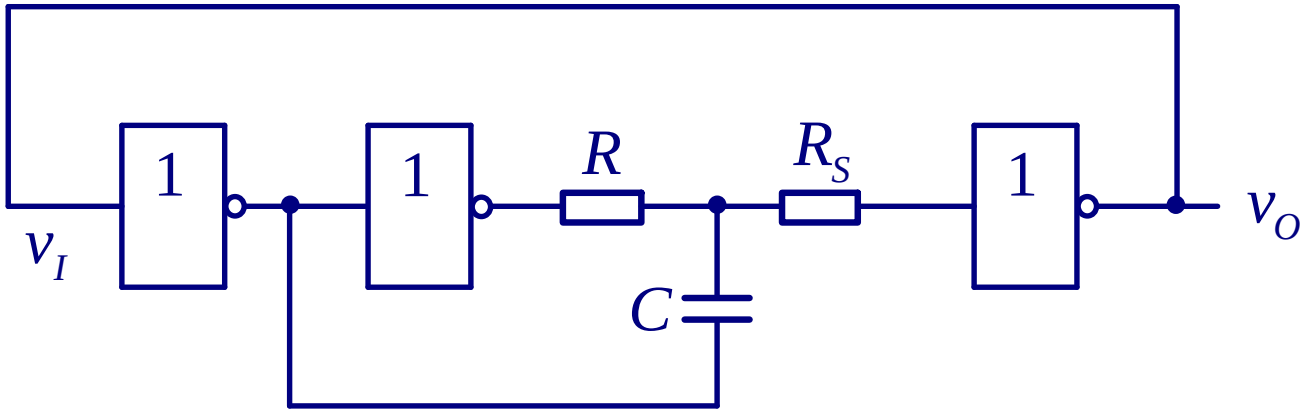
这种多谐振荡器结构简单，但不实用，原因是振荡频率太高，且不易调节

B) RC环形多谐振荡器

原理电路



实用电路

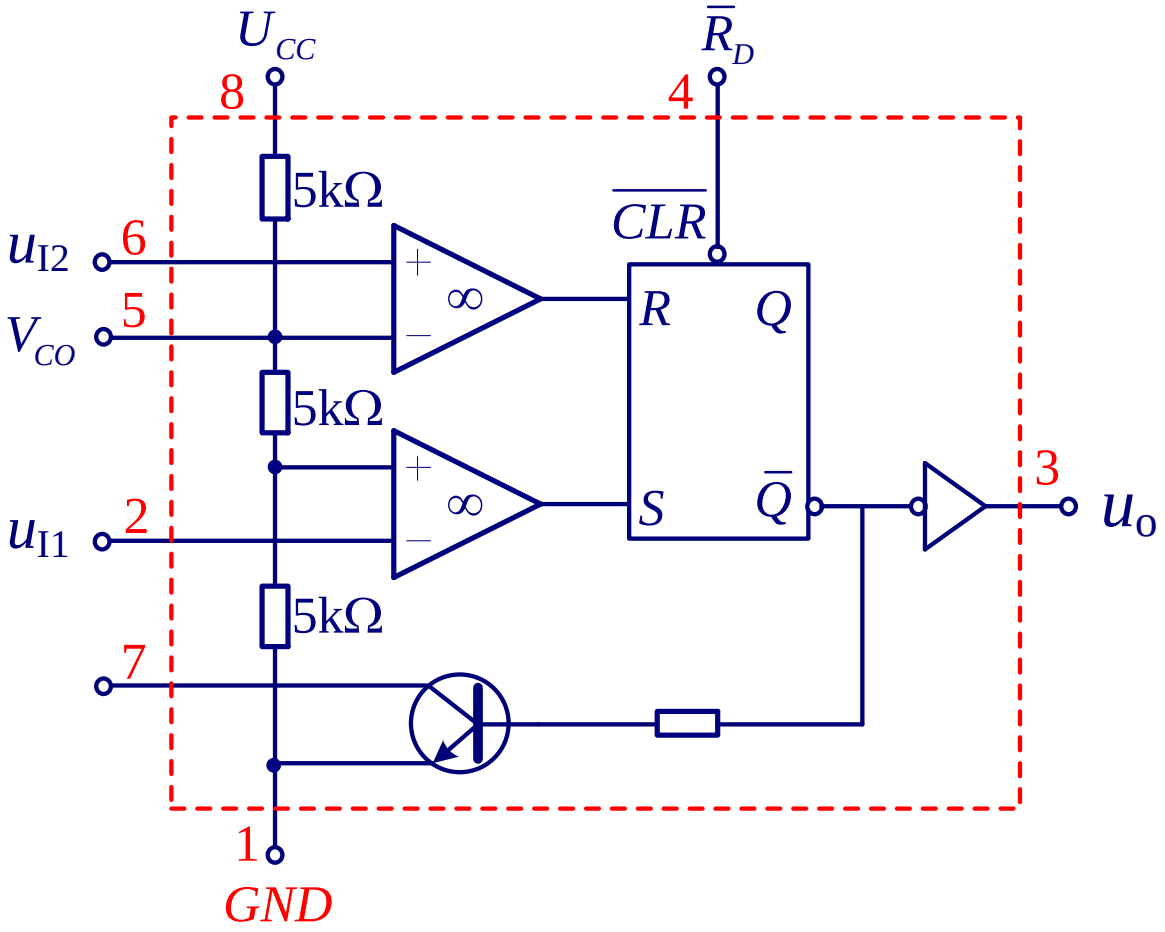


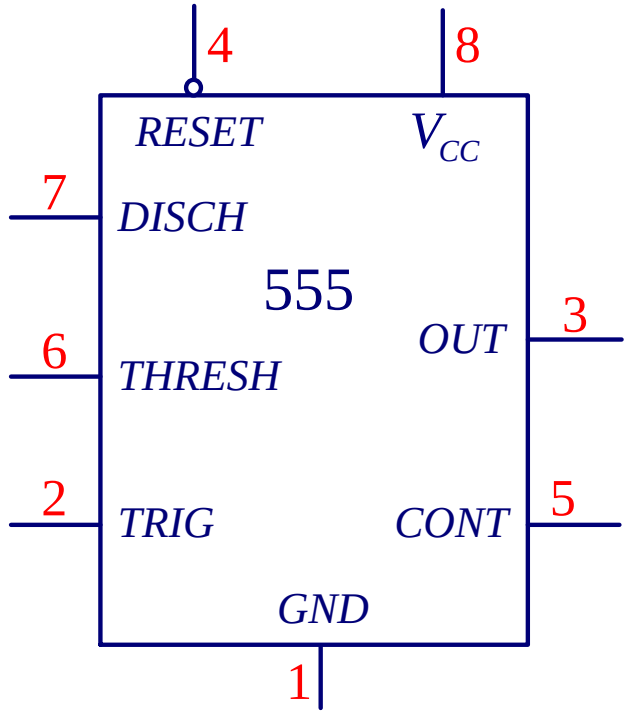
4、555定时器及其应用

555定时器是一种将模拟电路和数字电路集成于一体的电子器件。用它可以构成施密特触发器、单稳态触发器、多谐振荡器等多种电路。 555定时器在工业控制、定时、检测、报警等方面有广泛应用。

目前生产的定时器有双极型和CMOS两种类型，它们的结构及工作原理基本相同。双极型定时器具有较大的驱动能力， CMOS定时器则具有低功耗、输入阻抗高等优点。

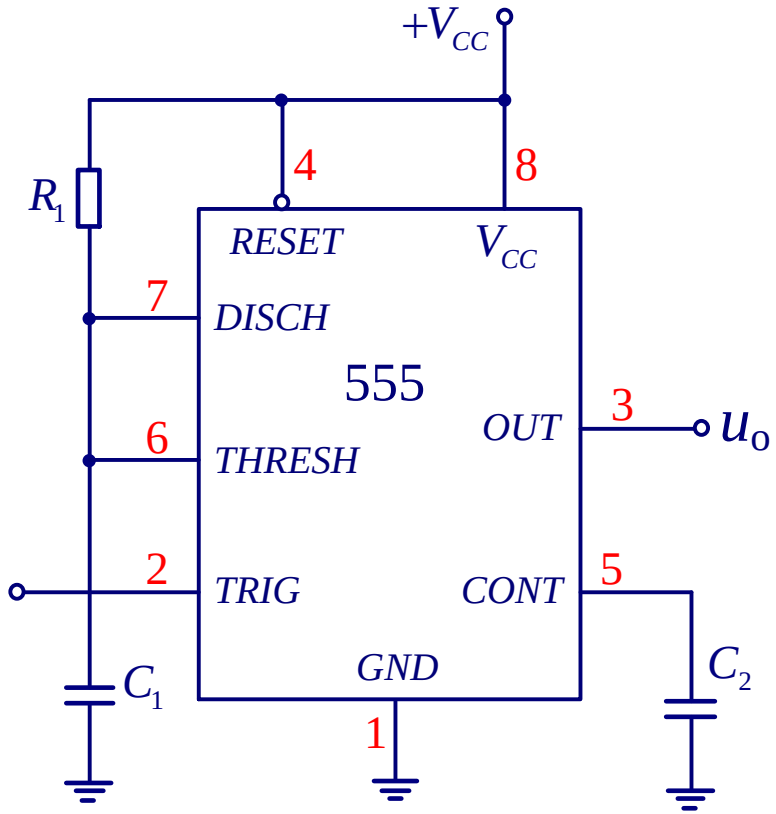
NE555

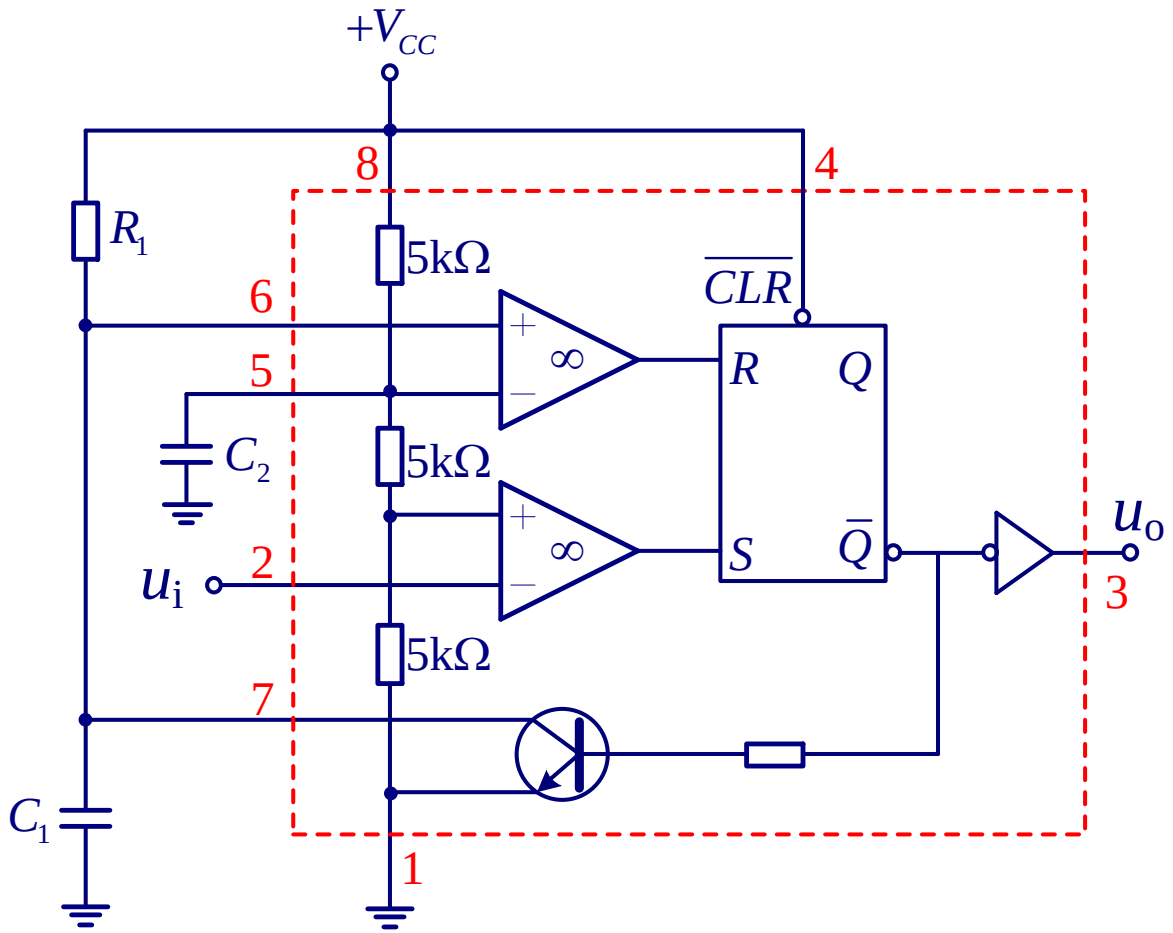

U_{CC}
8
$\overline{R}_D$
4
5kΩ
$\overline{CLR}$
u_{I2}
6
V_{CO}
5
+
−
∞
R
Q
5kΩ
S
$\overline{Q}$
3
u_o
u_{I1}
2
5kΩ
7
1
GND

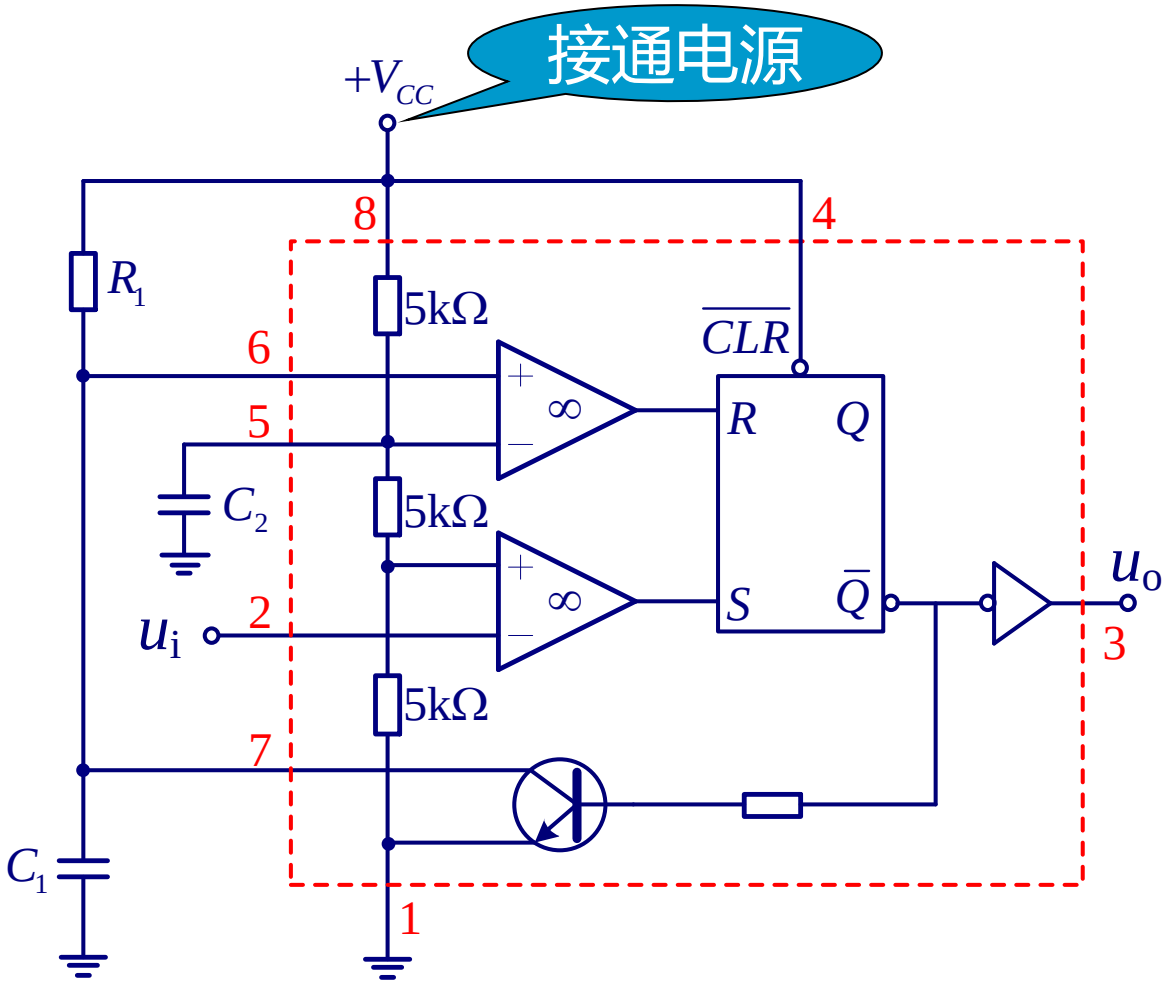

4
8
RESET
V_{CC}
7
DISCH
555
OUT
3
6
THRESH
2
TRIG
CONT
5
GND
1

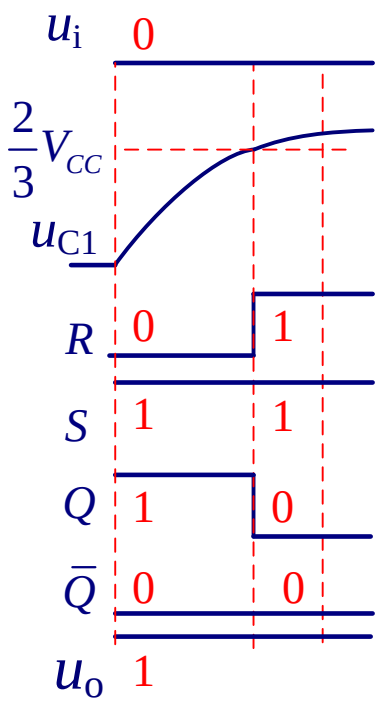
定时器电路的应用

1. 单稳态触发器



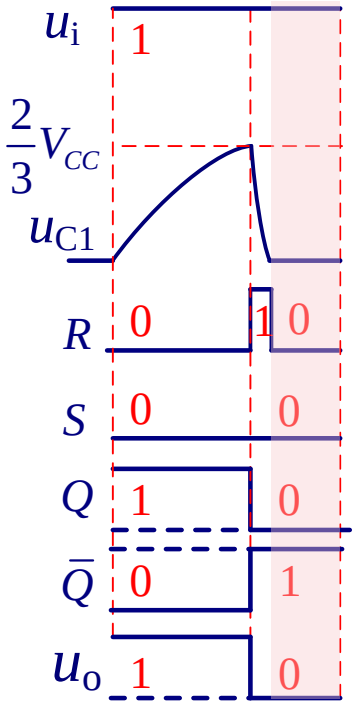




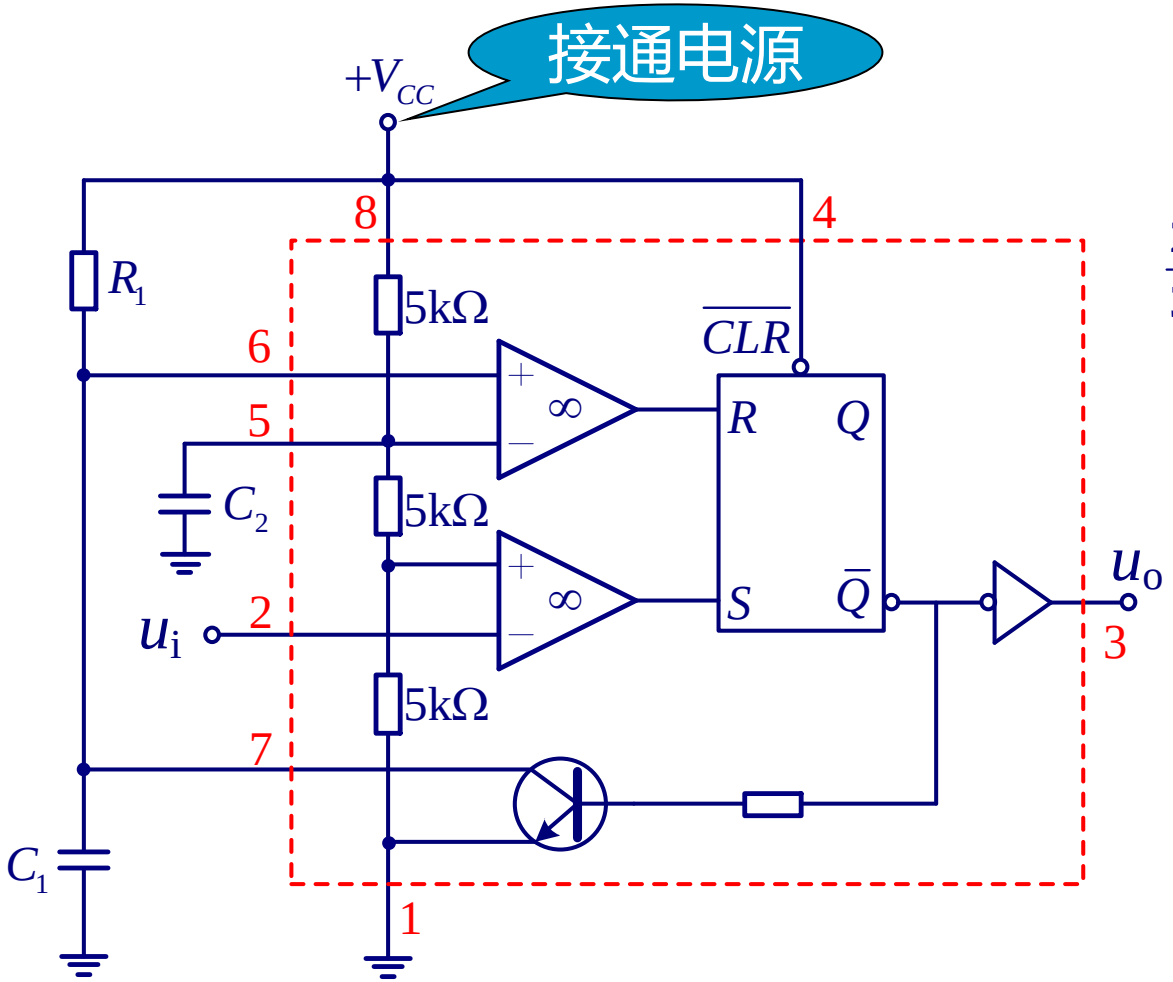


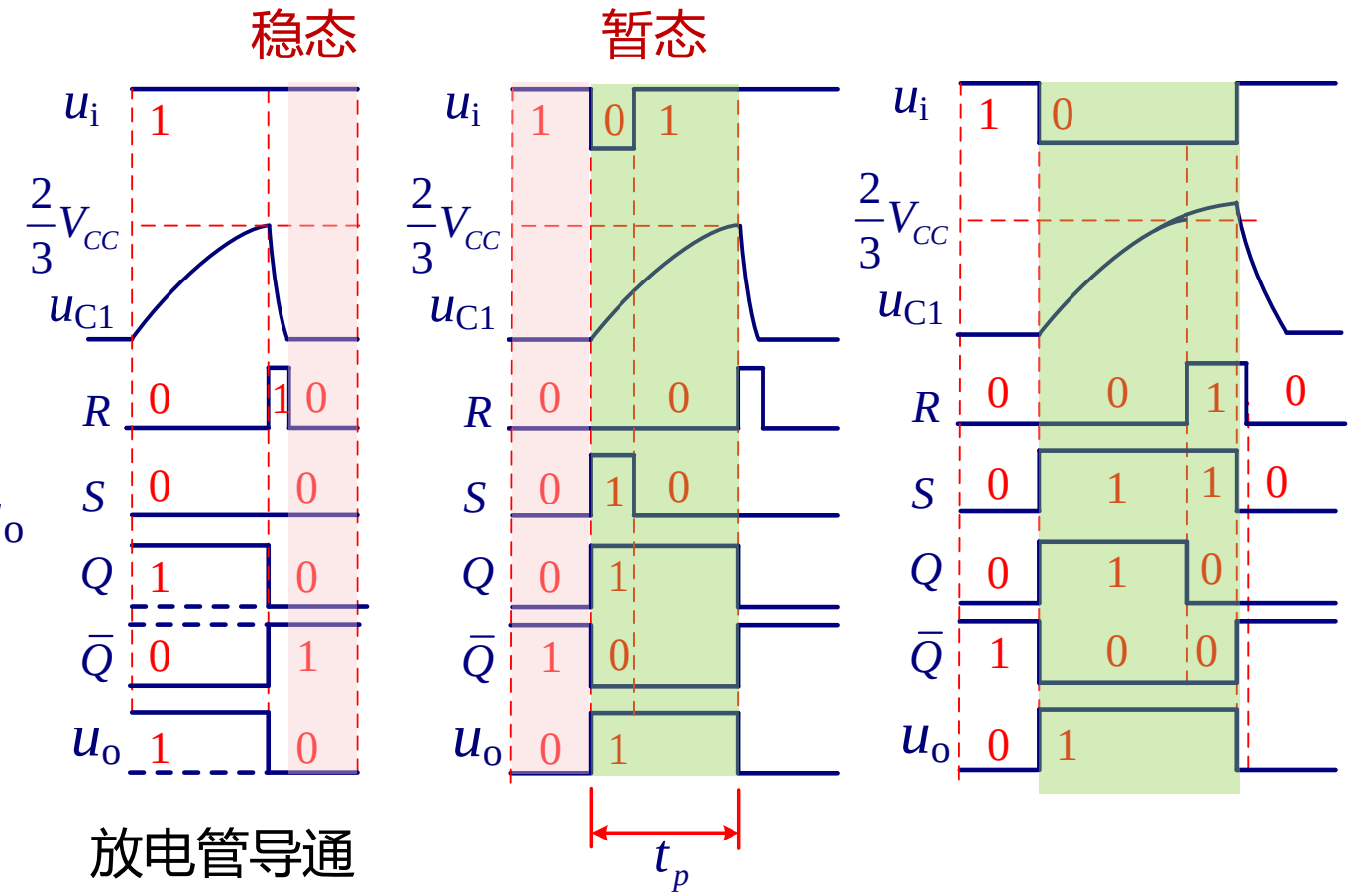
放电管截止

稳态



放电管导通





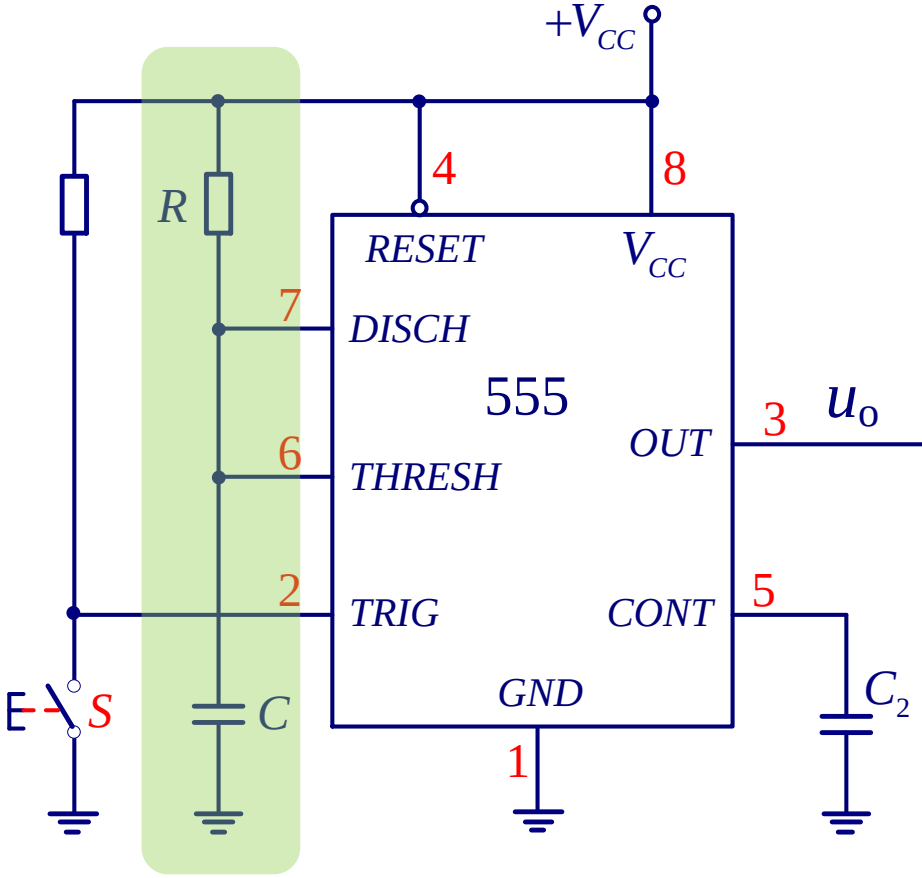
零状态响应

$$u_{C1}(t) = V_{CC} + (0 - V_{CC})e^{-\frac{t}{R_1C_1}}$$

$$\therefore \frac{2V_{CC}}{3} = V_{CC}(1 - e^{-\frac{t_p}{R_1C_1}})$$

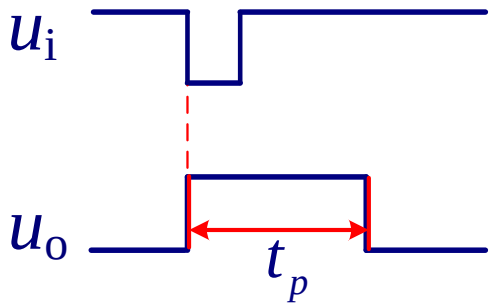
$$t_p = R_1C_1 \ln 3 \approx 1.1R_1C_1$$

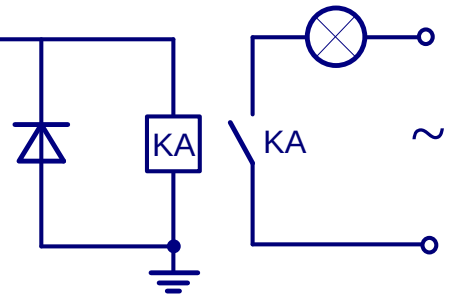
例：单稳态触发器构成短时用照明灯



若S未按下, 则 $u_i = 1$

若S按下, 则 $u_i = 0$



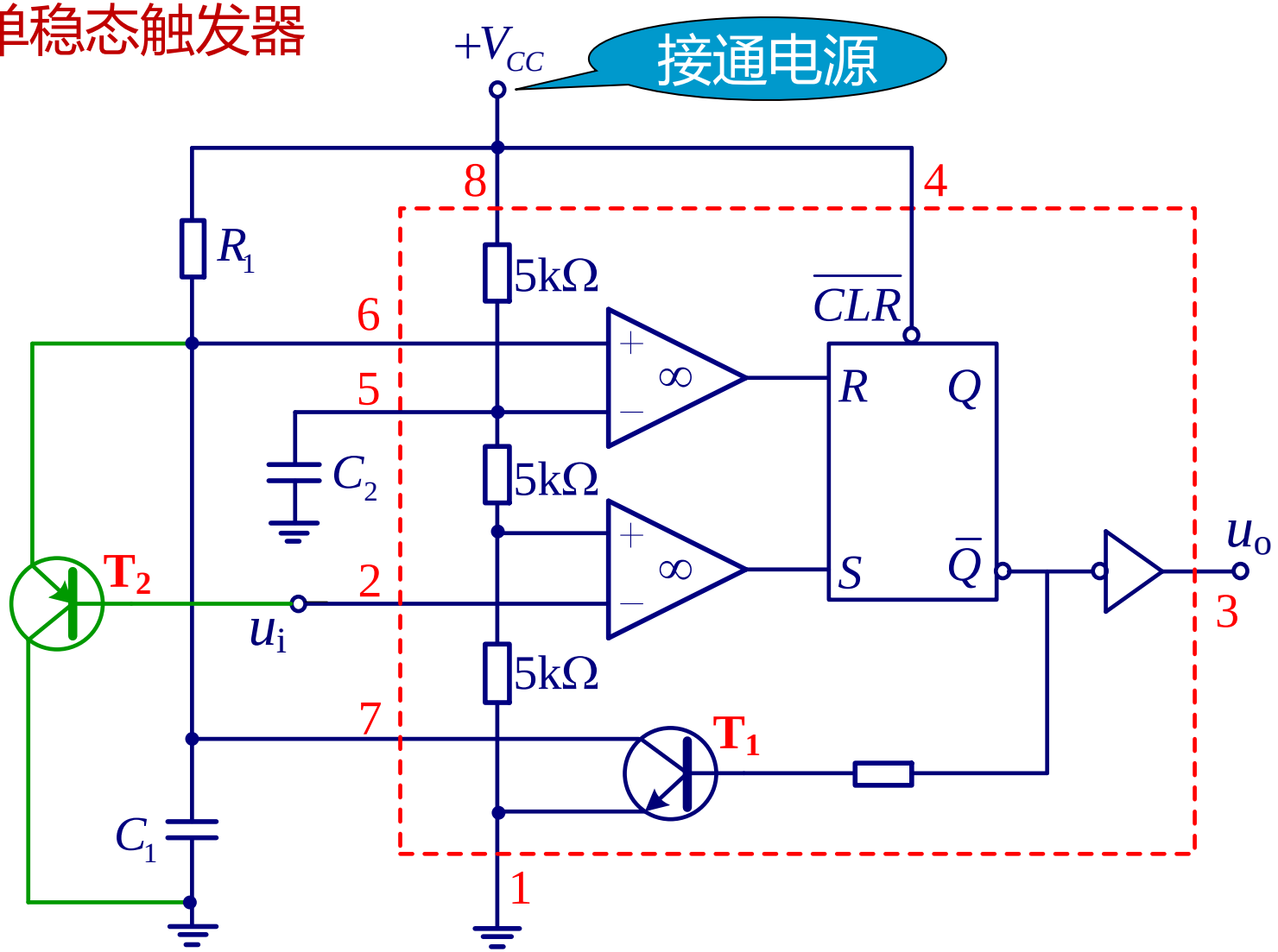


灯亮的时间为：

$$t_p = RC\ln 3 \approx 1.1RC$$

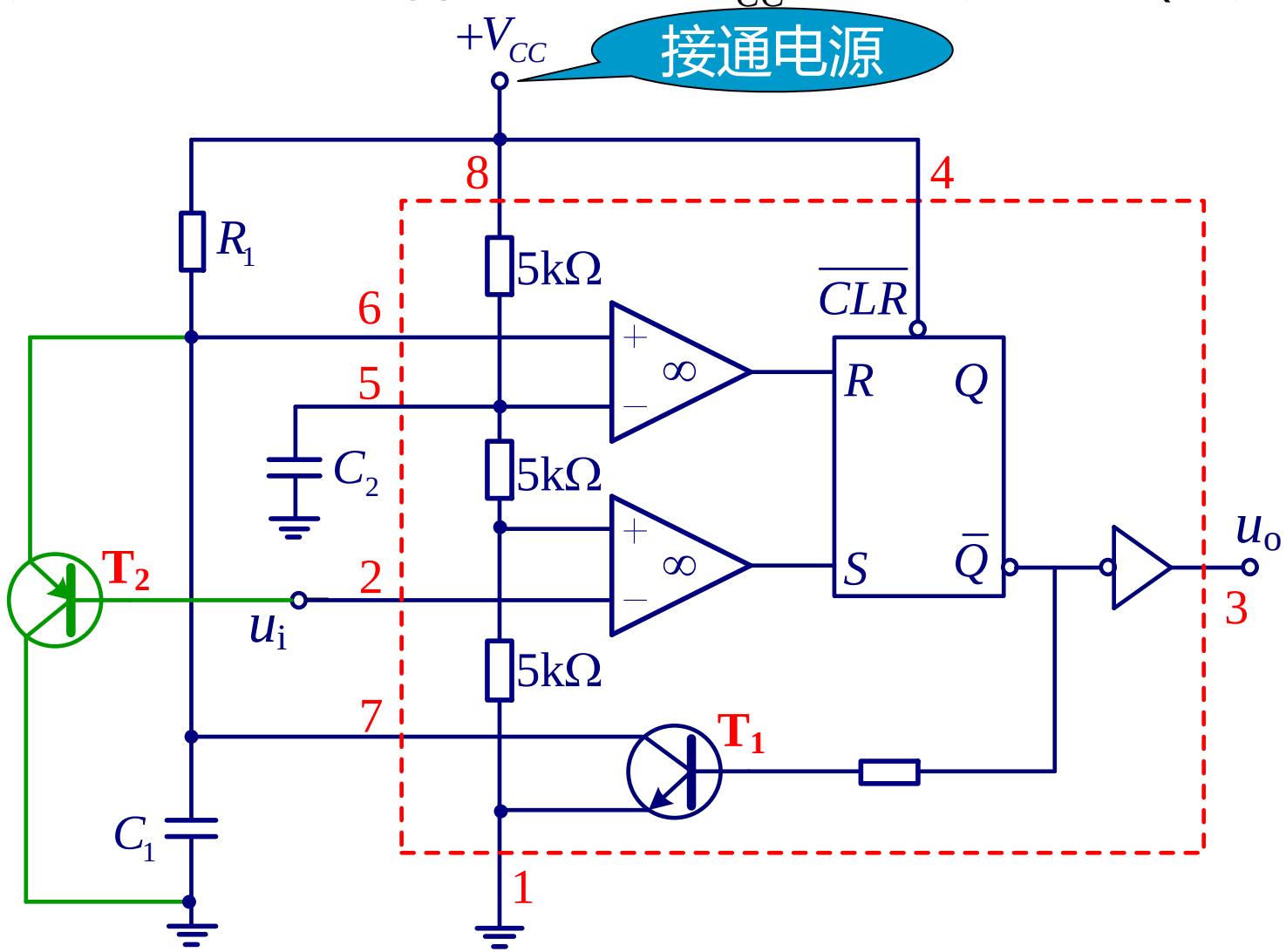
按钮 S	u_o	KA的线圈	KA 的触点	灯
未按下	0	不通电	断开	灭
按一下	1	通电	闭合	亮

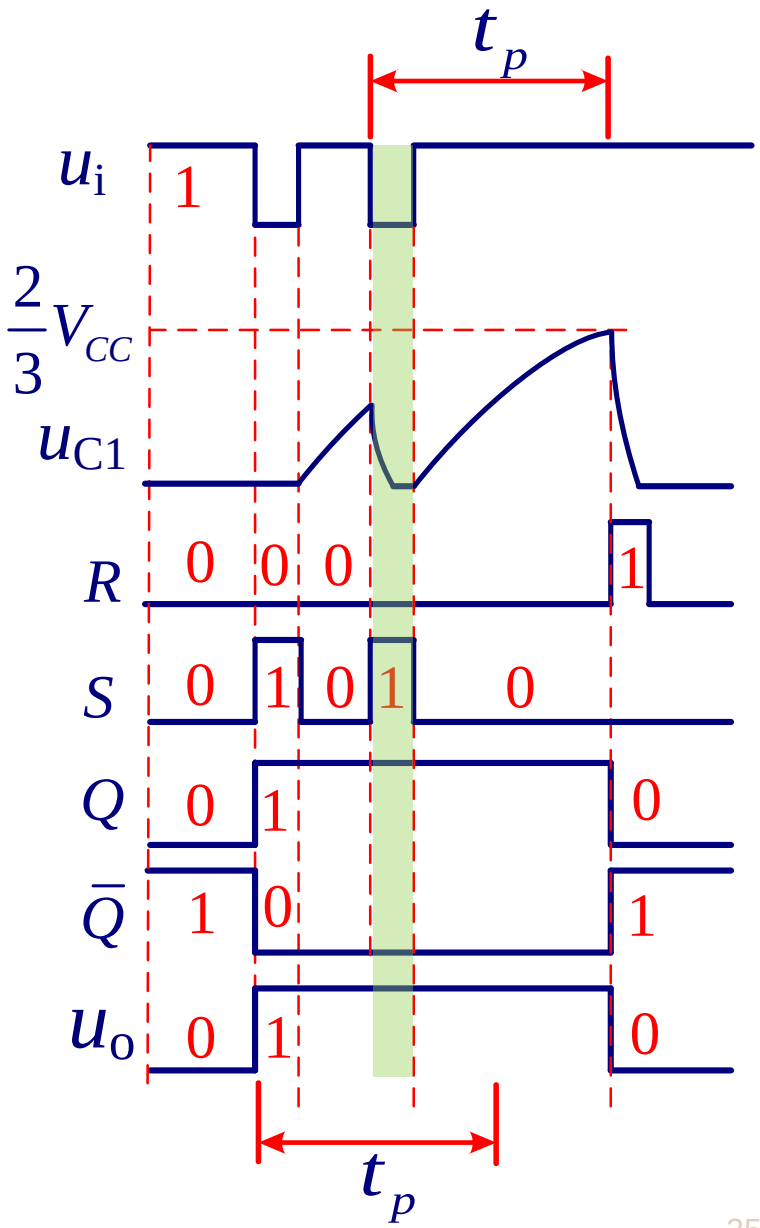
可重触发的单稳态触发器



T_1 高电平导通，T_2 低电平导通

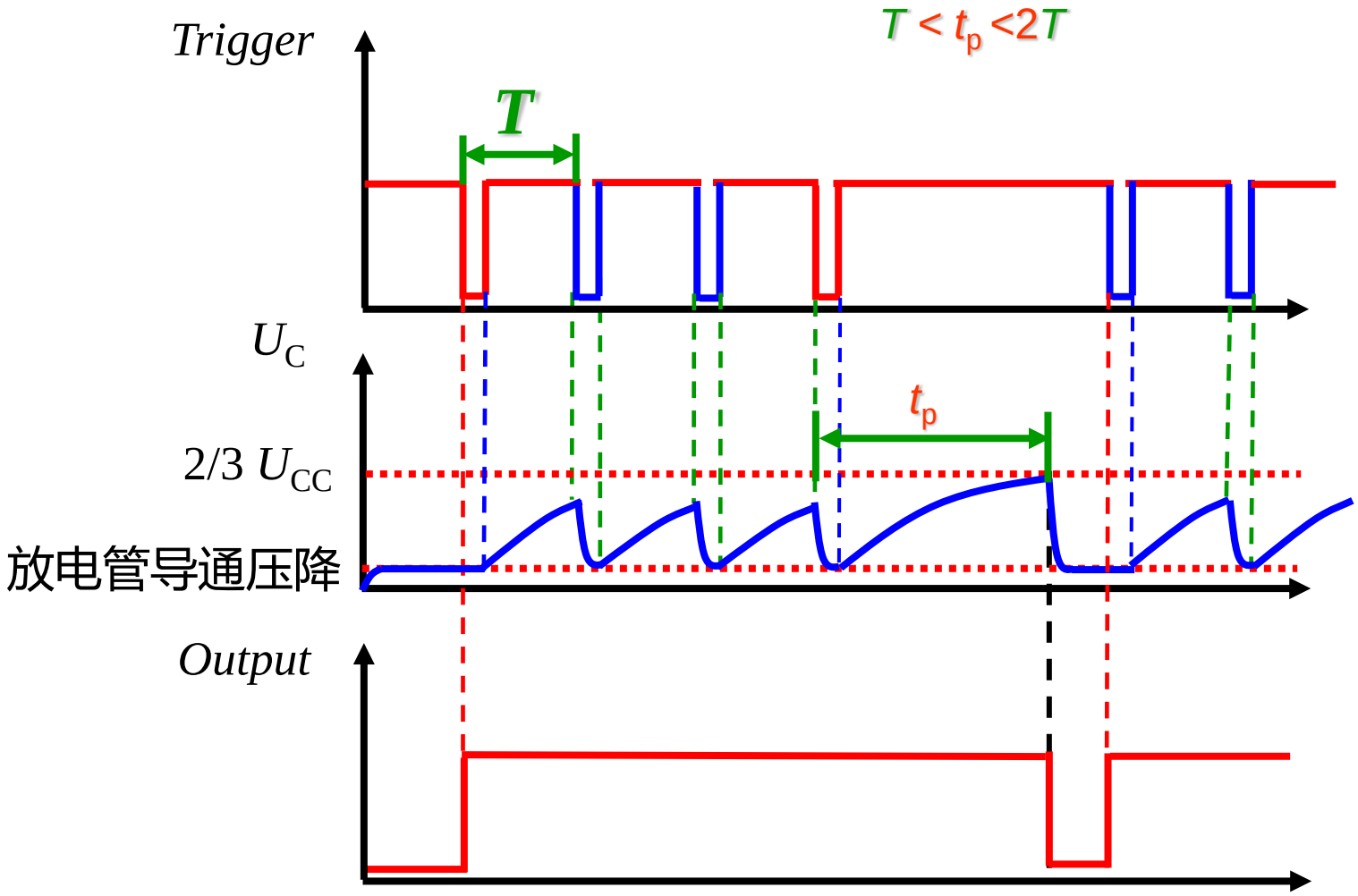
若触发窄脉冲在电容充电至$2/3U_{CC}$之前再次出现（再触发）



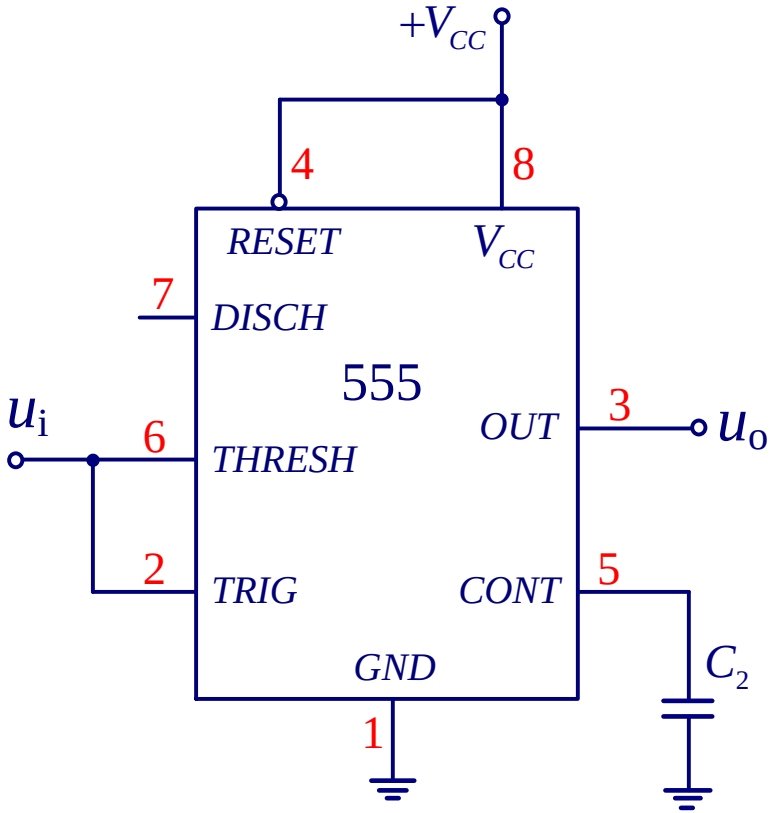


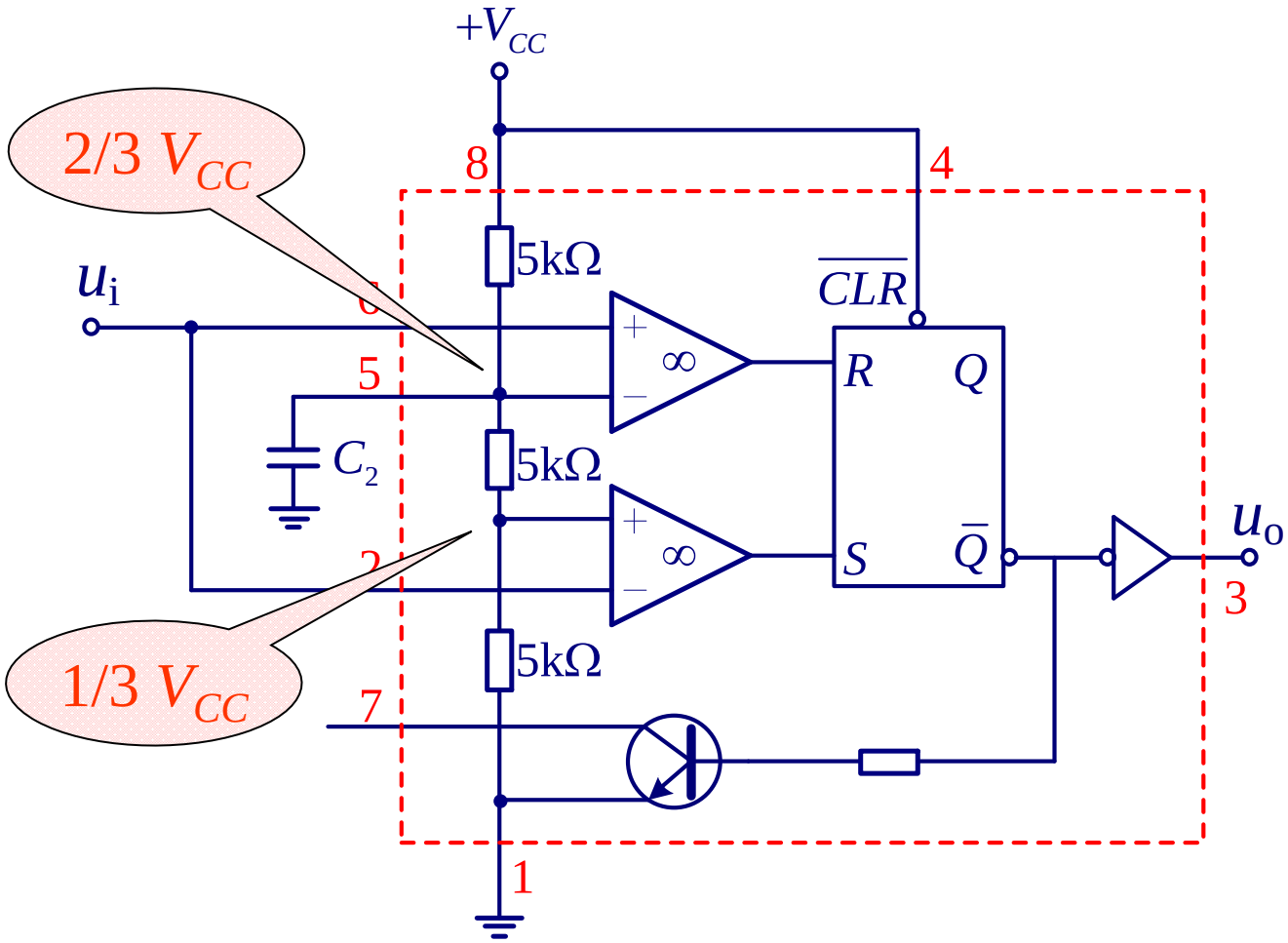
T_1 高电平导通，T_2 低电平导通

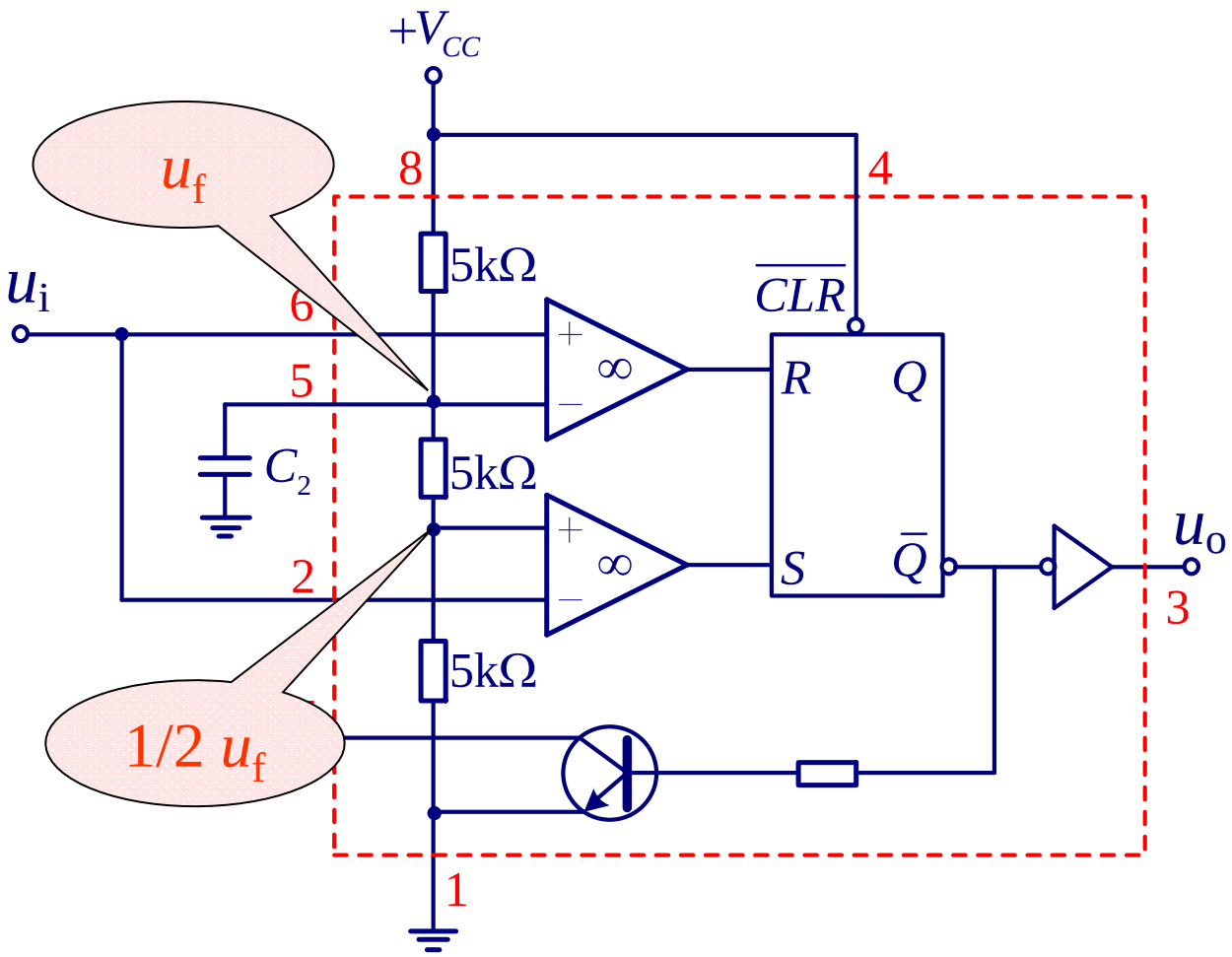
Application: Missing Pulse Detector

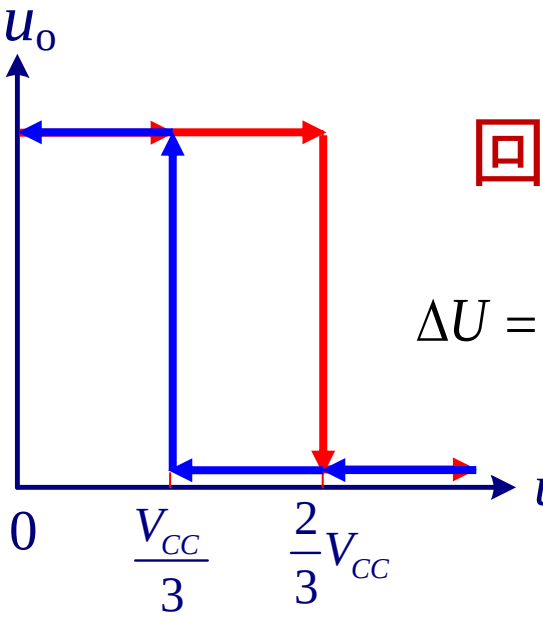


2. 施密特触发器


+VCC
4
8
RESET
VCC
7
DISCH
555
ui
6
THRESH
OUT
3
uo
2
TRIG
CONT
5
GND
C2
1

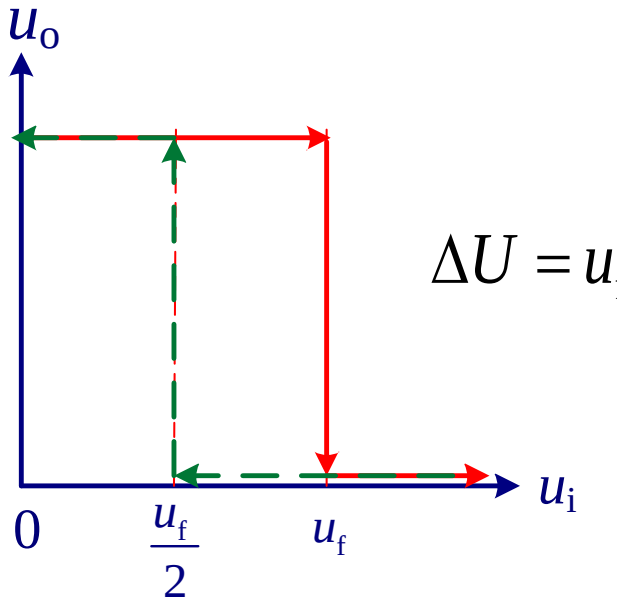

+VCC
2/3 VCC
8
4
5kΩ
ui
6
CLR
5
R
Q
C2
5kΩ
2
S
Q
uo
3
1/3 VCC
5kΩ
7
1





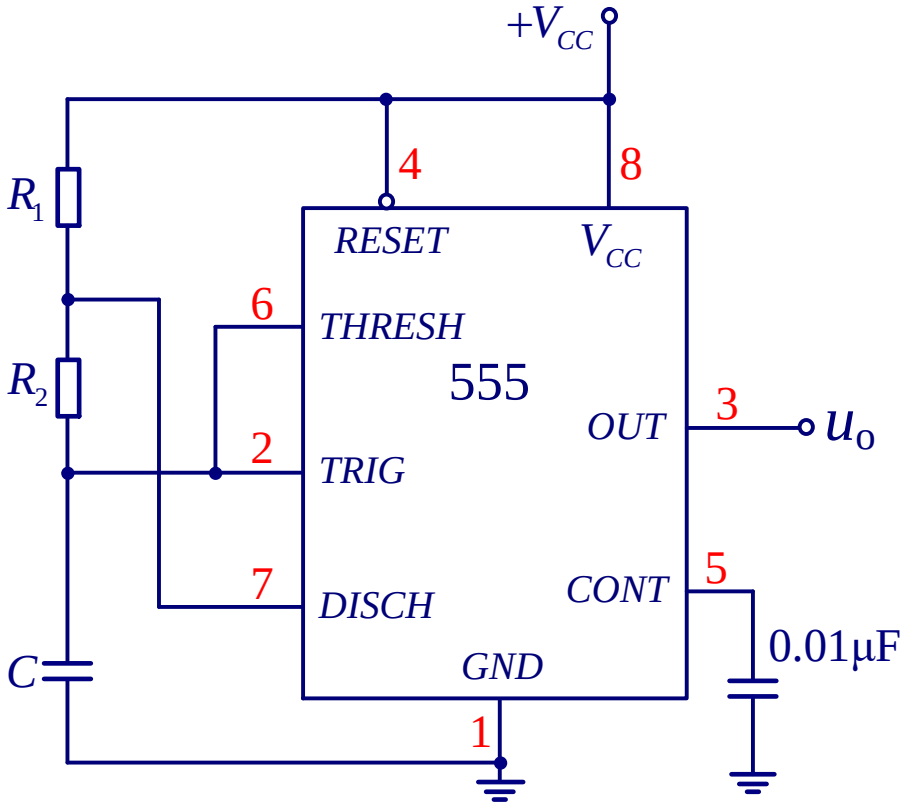
回差电压：

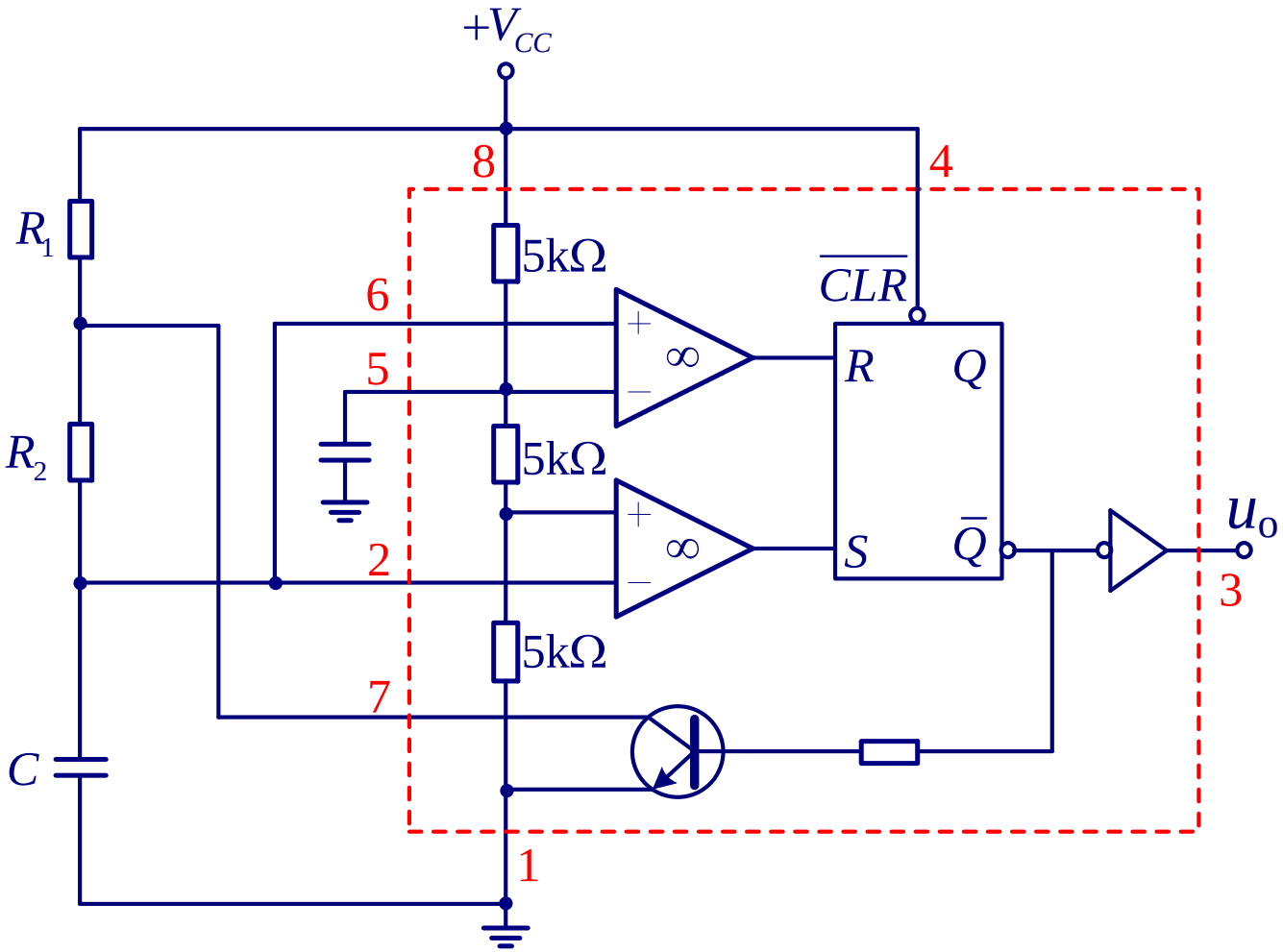
$$\Delta U = \frac{2}{3}V_{CC} - \frac{1}{3}V_{CC} = \frac{1}{3}V_{CC}$$

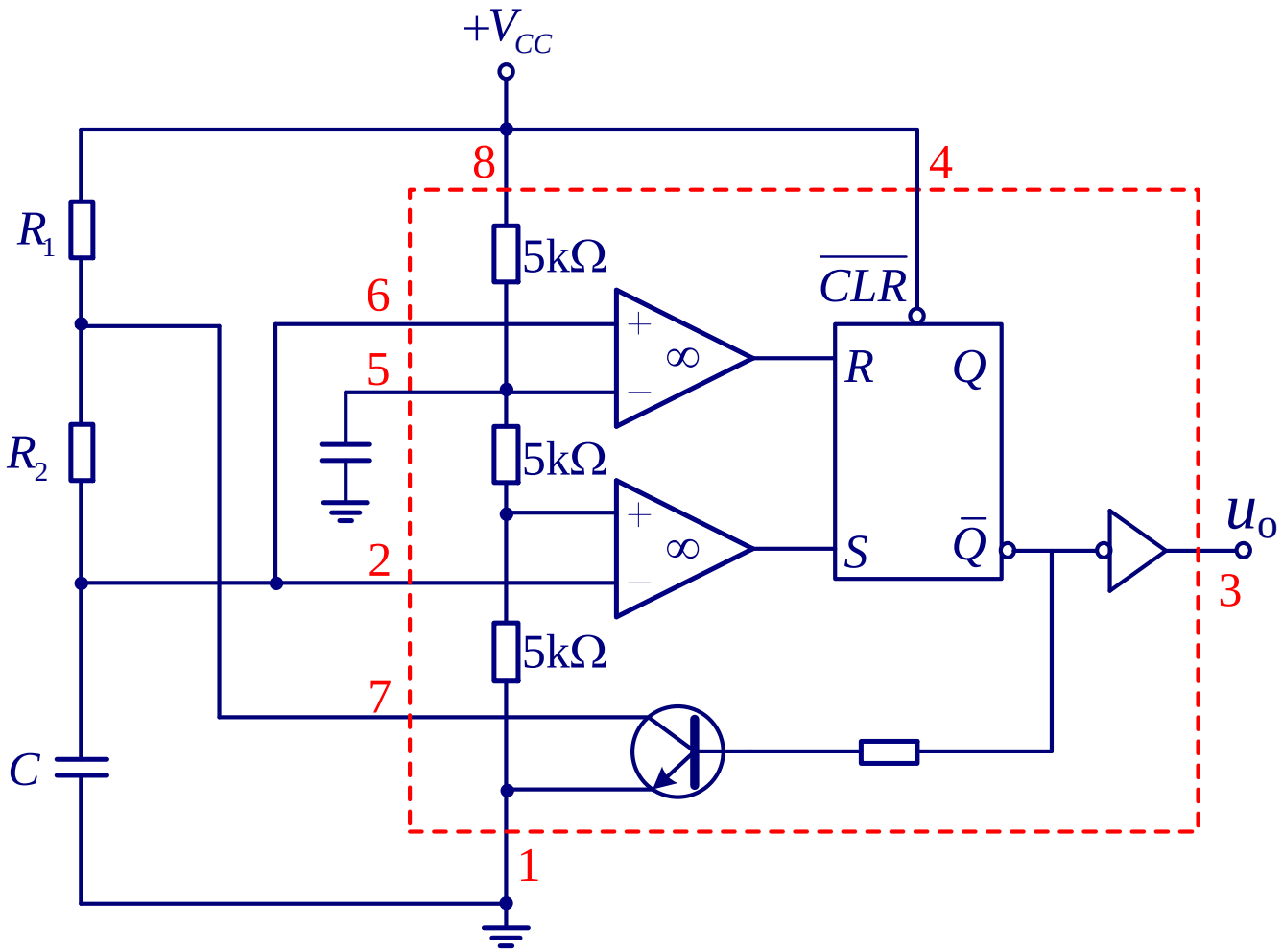


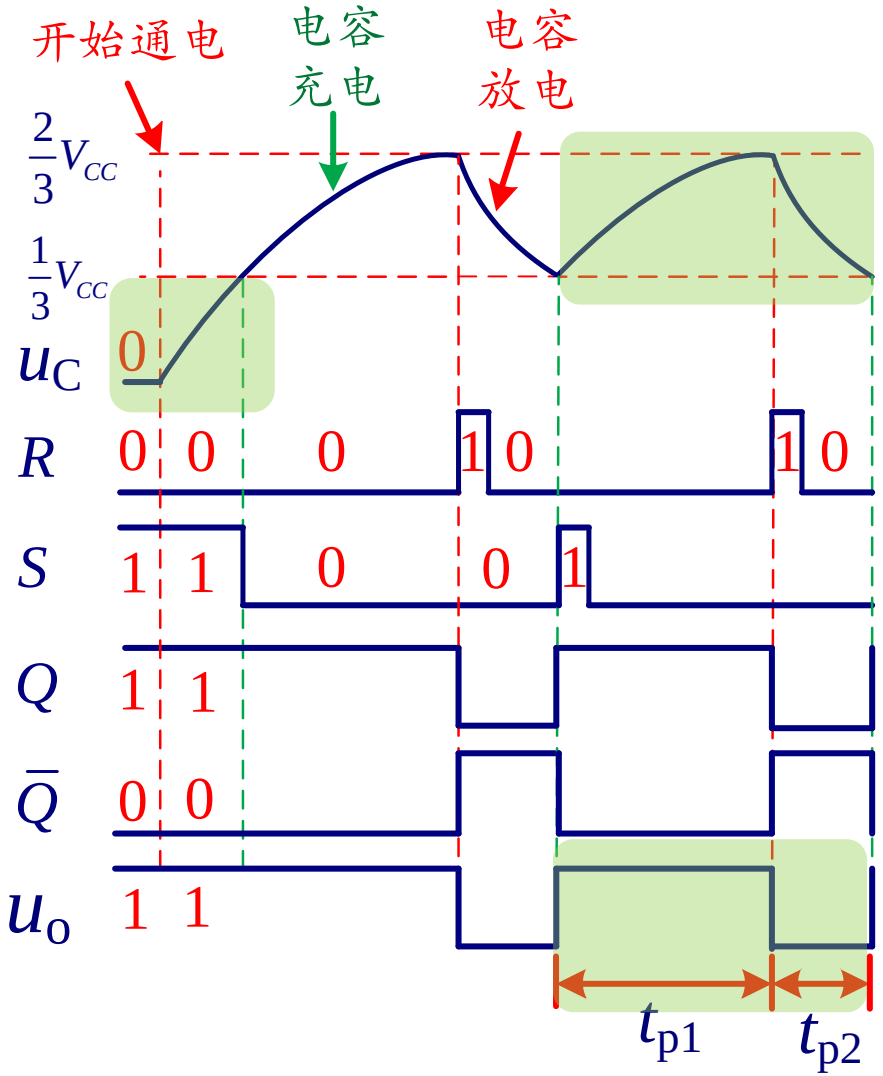
$$\Delta U = u_f - \frac{1}{2}u_f = \frac{1}{2}u_f$$

3. 多谐振荡器


+VCC
R1
R2
C
4
8
RESET
VCC
6
THRESH
555
2
TRIG
OUT
3
uo
7
DISCH
CONT
5
0.01μF
GND
1


+VCC
R1
R2
C
8
4
5kΩ
6
5
5kΩ
2
5kΩ
7
1
CLR
R
Q
S
Q
uo
3



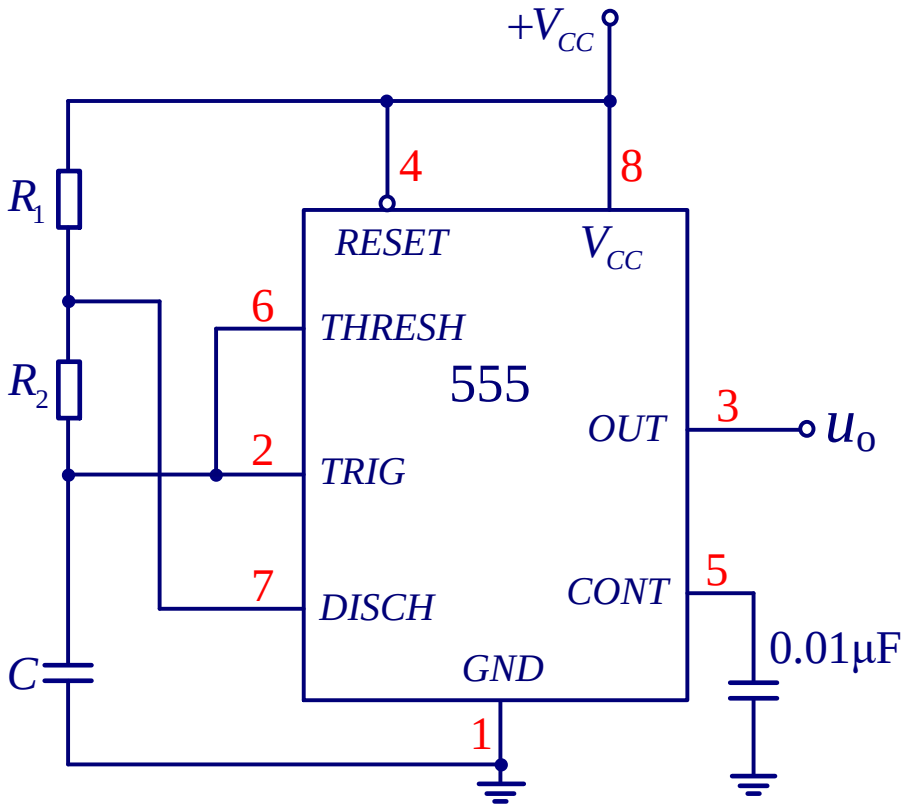


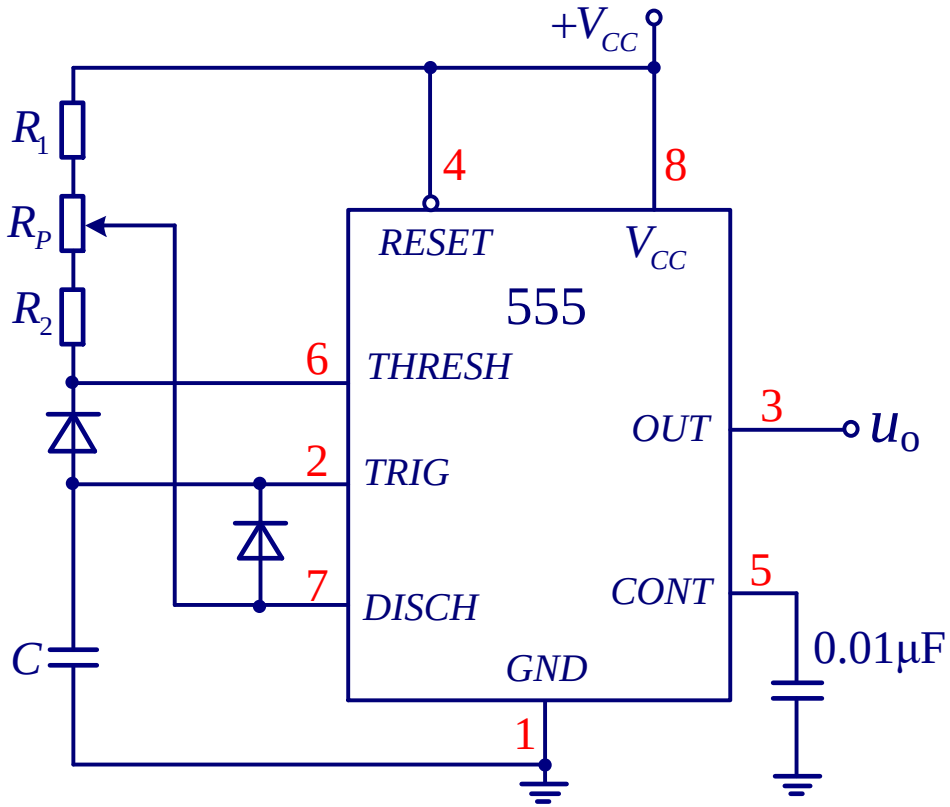
$$t_{p1} = (R_1 + R_2)C\ln 2 \approx 0.7(R_1 + R_2)C$$

$$t_{p2} = R_2 C\ln 2 \approx 0.7R_2 C$$

$$T = t_{p1} + t_{p2} \approx 0.7(R_1 + 2R_2)C$$

$$D = \frac{t_{p1}}{T} = \frac{R_1 + R_2}{R_1 + 2R_2}$$

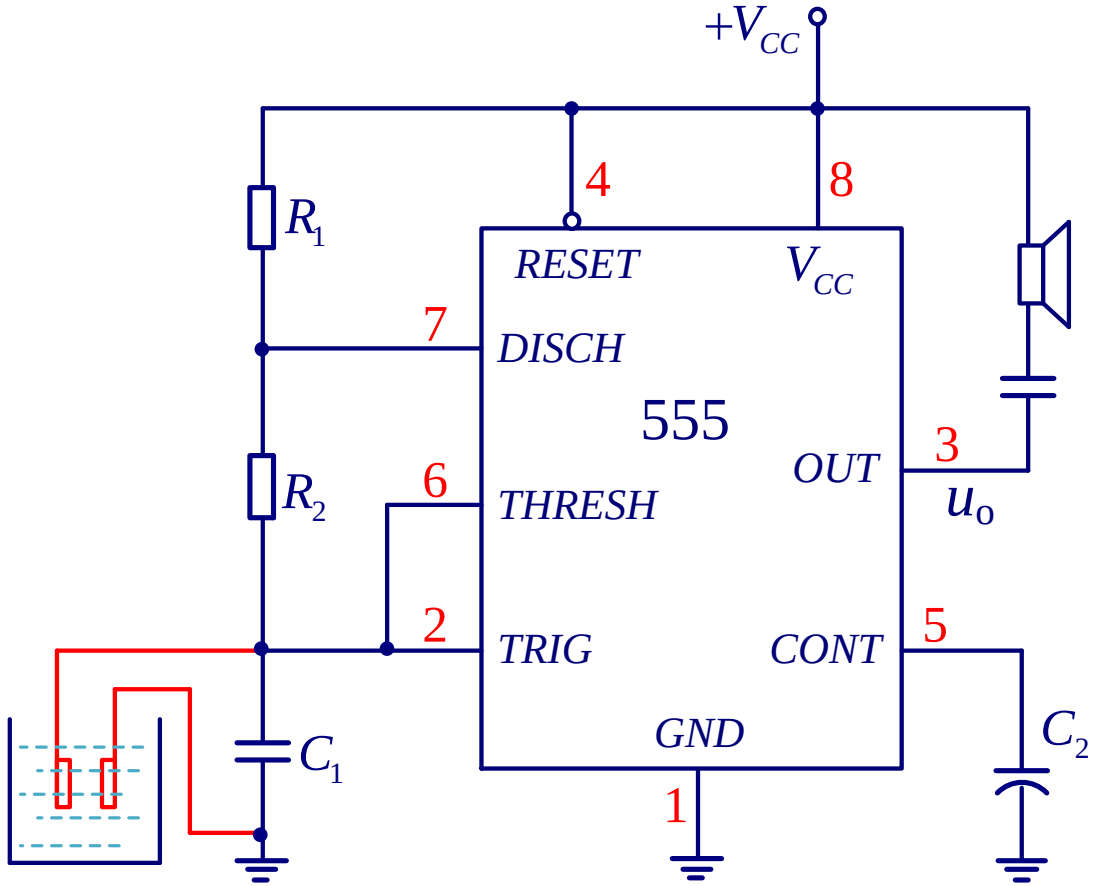




占空比：$D=\dfrac{t_{p1}}{T}=\dfrac{R_1+R_2}{R_1+2R_2}$

占空比：$D=\dfrac{t_{p1}}{T}=\dfrac{R_1'}{R_1'+R_2'}$

例：多谐振荡器构成水位监控报警电路



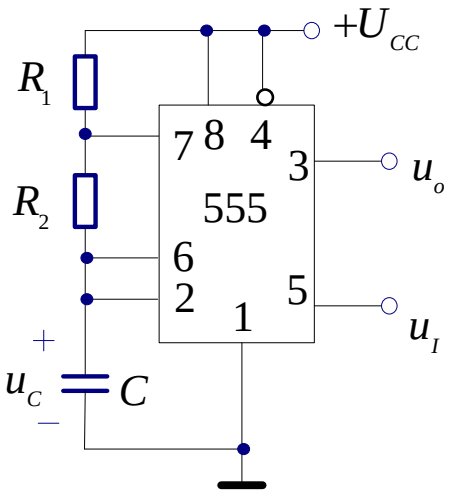
水位正常情况下，电容C被短接，扬声器不发声；水位下降到探测器以下时，多谐振荡器开始工作，扬声器发出报警。

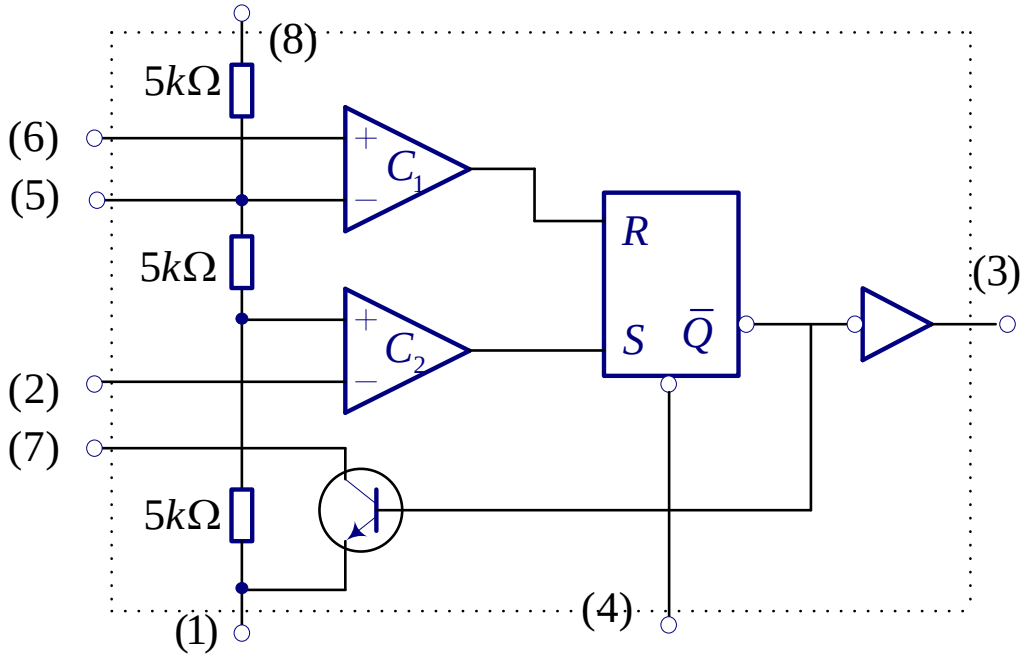
A 555 timer is configured as follow. $0<u_I<U_{CC}$

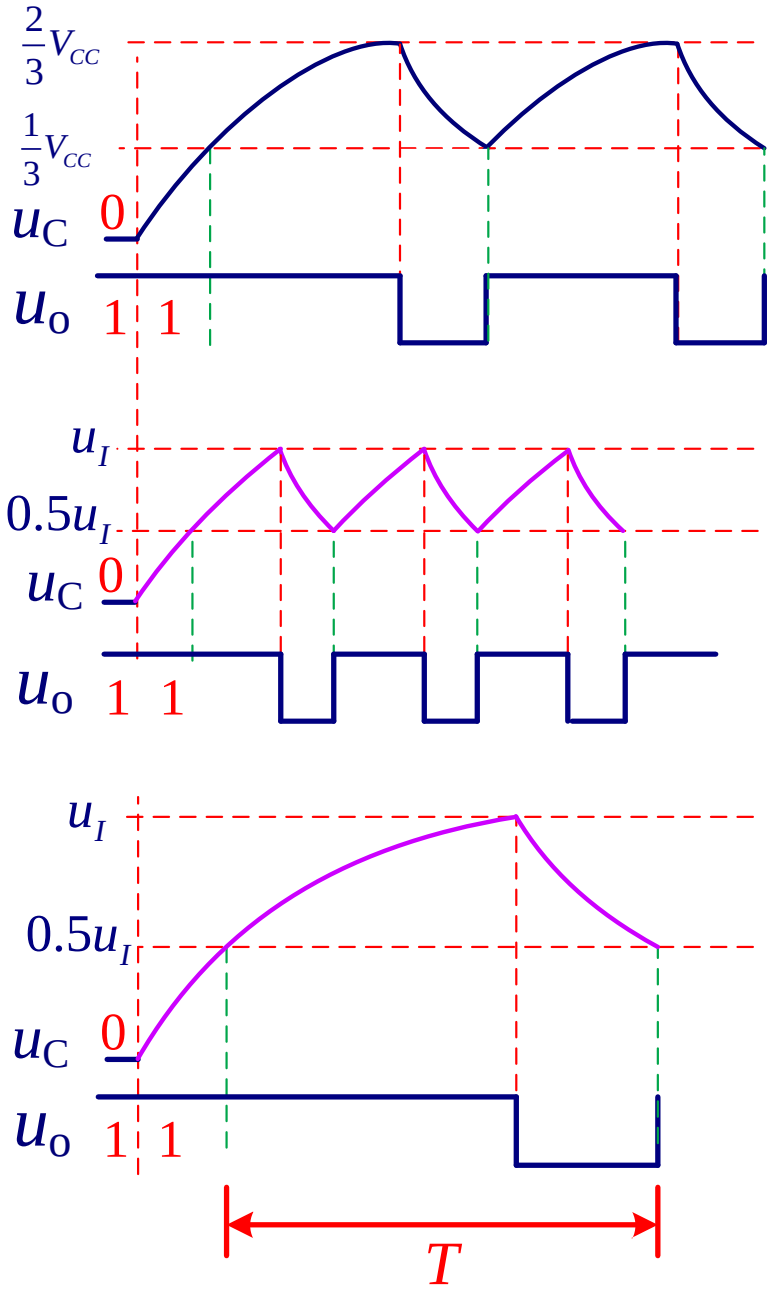
(1) Describe the operation of the circuit.

(2) Determine the frequency and the duty cycle for the output u_o

(3) Show the waveforms of u_C and u_o, and describe the relationship between the frequency of the output u_o and the input u_I







9-15　　D3~D0:0000

